

Minds with a Mission: The Role of Personality in a Multi-Agent System

1 Final Decision Maker Comparison Against State-of-the-Art and Human Performance

In this section, we statistically compare our approach against the state-of-the-art (SoTA) method for each dataset. A straightforward way to perform a single, overall significance test comparing the Final Decision Maker’s accuracy to the reported SoTA is to compute the weighted overall accuracy across all frameworks and then perform a binomial-proportion (z) test against the SoTA scores.

Because different frameworks were tested on different numbers of samples, we first aggregate our results by summing the total number of correct answers and the total number of items tested across all frameworks. Next, we convert each accuracy to a proportion by multiplying it by the corresponding number of samples, and then we sum these values to obtain the overall number of correct answers. Dividing this sum by the total number of samples across all frameworks gives us the overall weighted accuracy.

The null hypothesis is: H_0 : *There is no difference between the overall weighted accuracy of our Final Decision Maker and the state-of-the-art accuracy.*

To compare against human performance, we computed the confidence interval of our weighted results.

1.1 Strategy QA Dataset

We report in Table 1 the outcomes of the Final Decision Maker. We want to test whether our model’s true accuracy, denoted by p , differs from the state-of-the-art baseline, $p_0 = 0.72$. We conducted $N = 20,814$ trials, out of which we observed an overall accuracy of $\hat{p} = 0.7865$. Thus, we define the hypotheses as: $H_0 : p = 0.72$ and $H_1 : p \neq 0.72$. Under the null hypothesis

Framework	Accuracy (%)	Samples
Big Five Personality Traits	78.14	1560
Cognitive Behavioral Theory	78.41	1598
Cognitive Load Theory	79.40	1558
Dual-Process Theory	78.33	1546
Enneagram of Personality Traits	77.97	1593
Erikson’s Psychosocial Stages	79.23	1666
Flow Theory	78.77	1639
Freudian Psychoanalysis	79.10	1603
Mental Models	79.28	1578
Myers-Briggs Type Indicator	78.58	1601
Social Cognitive Theory	77.85	1612
User Design Persona	78.54	1631
User-Centered Design	78.51	1629

Table 1: Per-framework performance on the *Strategy QA* dataset.

H_0 , the standard error of the proportion is approximately 0.0031. The test statistic is $Z \approx 21.3$ with a p -value $< 10^{-90}$, and thus we *reject* the null hypothesis. Our model’s true accuracy is significantly higher than the 72% state-of-the-art; specifically, the point estimate is about 78.65%, which is a +6.65% improvement over the state-of-the-art. Statistically, the difference is extremely unlikely to be due to chance alone.

Our aggregated results yield an observed accuracy of $\hat{p} = 0.7865$ and a standard error (SE) of approximately 0.00284. A 95% confidence interval for the true accuracy is between 0.7809 and 0.7921. This indicates that if we were to repeat this evaluation on different samples of the same size from the dataset, we would expect the true accuracy of the Final Decision Maker to lie within this range 95% of the time. The narrow width of the interval reflects the large sample size and suggests that our estimate of approximately 78.65% accuracy is both precise and robust when generalizing to the entire dataset.

1.2 Common Sense QA Dataset

We report in Table 2 the performance of the Final Decision Maker across 13 different description frameworks. We want to test whether our model’s true

Framework	Accuracy (%)	Samples
Big Five Personality Traits	83.66	557
Cognitive Behavioral Theory	81.61	571
Cognitive Load Theory	83.75	560
Dual-Process Theory	81.68	546
Enneagram of Personality Traits	84.01	563
Erikson’s Psychosocial Stages	82.93	574
Flow Theory	82.75	545
Freudian Psychoanalysis	82.72	567
Mental Models	81.00	579
Myers-Briggs Type Indicator	82.61	575
Social Cognitive Theory	82.48	548
User Design Persona	84.47	573
User-Centered Design	82.28	553

Table 2: Per-framework performance on the *Common Sense QA* dataset.

accuracy p , differs from the state-of-the-art baseline, $p_0 = 0.912$. We conducted $N = 7311$ trials, from which we computed an aggregated (weighted) accuracy of $\hat{p} \approx 0.828$. Thus, we define the hypotheses as: $H_0 : p = 0.912$ and $H_1 : p \neq 0.912$. Under the null hypothesis H_0 , the standard error of the proportion is approximately $SE_{H_0} \approx 0.00331$. The test statistic is $Z \approx -25.38$, with a p -value $< 10^{-100}$, and thus we *reject* the null hypothesis. Our model’s true accuracy is significantly lower than the 91.20% state-of-the-art. To quantify the precision of our estimate, we calculate a 95% confidence interval for the model’s true accuracy using the observed proportion $\hat{p} = 0.828$. The standard error is computed as $SE \approx 0.00442$, yielding a 95% confidence interval between 0.819 and 0.837.

1.3 Last Letter Concat Dataset

We report in Table 3 the performance of the Final Decision Maker. We want to test whether our model’s true accuracy, denoted by p , differs from SoTA. From the aggregated data (weighted by the number of samples), we compute an accuracy of $\hat{p} \approx 0.9086$.

We first compare our aggregated accuracy to the state-of-the-art baseline, $p_0 = 0.86$. Thus, we define the hypotheses as: $H_0 : p = 0.86$ and $H_1 : p \neq$

Framework	Accuracy (%)	Samples
Big Five Personality Traits	92.20	500
Cognitive Behavioral Theory	92.00	500
Cognitive Load Theory	88.80	500
Dual-Process Theory	93.60	500
Enneagram of Personality Traits	91.60	500
Erikson’s Psychosocial Stages	83.80	500
Flow Theory	89.00	500
Freudian Psychoanalysis	92.80	500
Mental Models	92.20	500
Myers-Briggs Type Indicator	92.60	500
Social Cognitive Theory	92.40	500
User Design Persona	92.60	500
User-Centered Design	87.60	500

Table 3: Per-framework performance on the *Last Letter Concat* dataset.

0.86. With $N = 6500$, we have $\hat{p} = 0.9086$ and $p_0 = 0.86$. Under H_0 , the standard error is $SE_{H_0} \approx 0.0043$. The test statistic is then $Z \approx 11.3$, which is extremely large. The corresponding p -value is effectively $\ll 10^{-20}$, and thus we *reject* H_0 . Our model’s accuracy is significantly higher than the 86% state-of-the-art.

We further compare our results to human performance, treating human performance as $p_0 = 1.0$, then under $H_0 : p = 1.0$ there is zero variance (since the probability of an error is 0). In this case, a standard Z -test is not defined; however, from a binomial perspective, if $p = 1.0$ then even a single error would have probability 0. Given that we observed 594 errors, the p -value under this scenario is trivially $< 10^{-100}$. Hence, our model’s accuracy is *significantly below* the 100% human-performance level.

To quantify the precision of our estimate, we compute a 95% confidence interval for the model’s true accuracy p . The standard error calculated is $SE \approx 0.00358$. This indicates that we can be 95% confident that the model’s true accuracy on this dataset lies between 90.16% and 91.56%.

In summary, when we aggregate results across all trials, the Final Decision Maker achieves an overall accuracy of about 90.86%. Statistical tests confirm that this accuracy is significantly higher than the 86% state-of-the-

art approach ($Z \approx 11.3$, $p \ll 10^{-20}$) and significantly lower than a hypothetical 100% human-performance. The 95% confidence interval further indicates that the model’s true performance is stable around approximately 91% when generalized to the entire dataset.

1.4 Social IQa Dataset

Framework	Accuracy (%)	Samples
Big Five Personality Traits	79.29	1700
Cognitive Behavioral Theory	79.47	1646
Cognitive Load Theory	78.89	1644
Dual-Process Theory	78.28	1625
Enneagram of Personality Traits	78.98	1670
Erikson’s Psychosocial Stages	78.70	1737
Flow Theory	79.02	1654
Freudian Psychoanalysis	79.09	1722
Mental Models	78.48	1710
Myers-Briggs Type Indicator	78.05	1695
Social Cognitive Theory	78.99	1623
User Design Persona	80.58	1715
User-Centered Design	78.45	1666

Table 4: Per-framework performance on the *Social IQa* dataset.

We report in Table 4 the performance of the Final Decision Maker. We want to test whether our model’s true accuracy, denoted by p , differs from established SoTA. From the aggregated data weighted by the number of samples, we compute an accuracy of $\hat{p} \approx 0.7891$.

We first compare our aggregated accuracy to the state-of-the-art baseline, $p_0 = 0.832$. Thus, we define the hypotheses as: $H_0 : p = 0.832$ and $H_1 : p \neq 0.832$. With $N = 21807$ trials, we have $\hat{p} = 0.7891$ and $p_0 = 0.832$. Under H_0 , the standard error $SE_{H_0} \approx 0.00253$. The test statistic is then $Z \approx -16.93$, with a p -value effectively close to zero ($p \ll 10^{-30}$). Thus, we *reject* H_0 , concluding that our model’s accuracy is significantly lower than the 83.20% state-of-the-art.

We further compare our results to a human performance of $p_0 = 0.87$. Here, the hypotheses are defined as: $H_0 : p = 0.87$ and $H_1 : p \neq 0.87$.

Under H_0 , the standard error $SE_{H_0} \approx 0.00228$. The resulting test statistic is $Z \approx -35.5$, yielding a small p -value of $\ll 10^{-200}$. We, therefore, *reject* H_0 , conclude that our model’s accuracy is significantly below the human performance of 87.00%.

To quantify the precision of our estimate, we calculate a 95% confidence interval for the model’s true accuracy p . The standard error $SE \approx 0.00276$, and the 95% confidence interval lies between 78.37% and 79.45%.

In summary, when aggregating results across all trials, the Final Decision Maker achieves an overall accuracy of about 78.91%. Statistical tests confirm that this performance is significantly lower than both the state-of-the-art and human performance.

1.5 Social Support Dataset

1.6 Accuracy

Since no state-of-the-art approaches have reported accuracy for this dataset, we only calculate the confidence interval for our approach. We treat these 13 outcomes (reported in Table 5) of the description frameworks as a sample of size $n = 13$. The sample mean is $\bar{X} = 41.69\%$, and the sample standard deviation is $s \approx 1.33\%$. To approximate the true mean performance μ of the method, we construct a 95% confidence interval using a t -based approach. With $n - 1 = 12$ degrees of freedom, the critical value is $t_{0.025, 12} \approx 2.18$. The standard error is $SE \approx 0.37\%$, yielding a margin of error of $2.18 \times 0.37\% \approx 0.80\%$. Thus, the 95% confidence interval for the true mean lies between 40.89% and 42.49%.

This relatively narrow interval reflects the modest spread of the sample, as indicated by a standard deviation of about 1.33%.

1.6.1 Macro F1

Data, Accuracy, and Macro-F1 Scores We report in Table 5 the Final Decision Maker’s macro-F1 scores across the 13 description frameworks. We want to test whether the average macro-F1 across these 13 runs (μ), differs from the state-of-the-art. From the data, we compute a sample mean of $\bar{X} = 27.60\%$ and a sample standard deviation of $s = 5.06\%$. Treating these 13 scores as independent observations, we perform a one-sample t -test with $n = 13$.

The hypotheses are defined as $H_0 : \mu = 29\%$ and $H_1 : \mu \neq 29\%$. The standard error is $SE \approx 1.40\%$. The test statistic $t \approx -1.00$. With $n - 1 =$

Framework	Accuracy (%)	Macro-F1 (%)	Samples
Big Five Personality Traits	40.02	22.86	897
Cognitive Behavioral Theory	43.48	30.34	897
Cognitive Load Theory	40.02	22.84	897
Dual-Process Theory	40.58	23.29	897
Enneagram of Personality Traits	42.59	29.85	897
Erikson’s Psychosocial Stages	40.25	22.98	897
Flow Theory	42.25	29.84	897
Freudian Psychoanalysis	41.36	29.34	897
Mental Models	41.58	29.35	897
Myers-Briggs Type Indicator	42.81	40.25	897
Social Cognitive Theory	43.48	24.33	897
User Design Persona	40.47	23.15	897
User-Centered Design	43.03	30.40	897

Table 5: Per-framework performance on the *Social Support* dataset.

12 degrees of freedom, this t -value corresponds to a two-sided p -value of approximately 0.34, which is not statistically significant.

We also construct a 95% confidence interval for the true mean macro-F1, that lies between 24.54% and 30.66%. Since 29% lies within this interval, we fail to reject the null hypothesis.

In summary, across the 13 expert-model configurations, the Final Decision Maker achieves an average macro-F1 of 27.60% (with individual scores ranging from about 22.8% to 40.3%). Although this average is slightly below the state-of-the-art approach of 29%, the one-sample t -test indicates that there is insufficient evidence to conclude that the mean performance is statistically different from 29%.

2 Description Framework Analyses

To understand which is the best-performing description framework, we performed a global analysis to compare the performance of different description frameworks based on both the Majority Vote outcome and the Final Decision Maker outcome.

For each framework, we aggregated the total number of successes and failures across all datasets and constructed a contingency table. A chi-square test was then applied to determine whether there were statistically significant differences in the outcomes performance among the frameworks. The chi-square test null hypothesis H_0 is: *there is no association between*

the description framework and the success/failure outcome of the ensembling method adopted, i.e., all frameworks perform equivalently.

2.1 Majority Vote Analysis

Framework	Successes	Failures
Big Five Personality Traits	3864.9936	1349.0064
Cognitive Behavioral Theory	3904.0689	1307.9311
Cognitive Load Theory	3827.0044	1331.9956
Dual-Process Theory	3784.0509	1329.9491
Enneagram of Personality Traits	3887.0043	1335.9957
Erikson’s Psychosocial Stages	3995.9253	1378.0747
Flow Theory	3915.9107	1319.0893
Freudian Psychoanalysis	3950.0357	1338.9643
Mental Models	3899.0131	1364.9869
Myers-Briggs Type Indicator	3904.1322	1363.8678
Social Cognitive Theory	3875.9580	1304.0420
User Design Persona	4002.9301	1313.0699
User-Centered Design	3888.0289	1356.9711

Table 6: Global Contingency Table of Majority Vote Successes and Failures by Framework.

The chi-square test yielded a statistic of 5.54 with 12 degrees of freedom, corresponding to a p -value of 0.9376. Since the p -value is very high, we fail to reject the null hypothesis. Thus, there is no statistically significant difference in the Majority Vote performance among the various description frameworks. Table 6 shows the global contingency table with the aggregated majority vote successes and failures for each description framework. As the table indicates, the number of successes and failures are very similar across the frameworks. Such close results are consistent across the frameworks and reinforce the conclusion from the chi-square test.

Thus, these results suggest that, when a majority vote ensembling method is used to aggregate decisions, the choice of description framework does not substantially affect the outcome. Consequently, factors other than majority vote performance, such as theoretical underpinnings, interpretability, or

computational efficiency, may be more critical when selecting a description framework for a given task or application.

2.2 Final Decision Maker Analysis

Framework	Successes	Failures
Big Five Personality Traits	3852.88	1361.12
Cognitive Behavioral Theory	3877.08	1334.92
Cognitive Load Theory	3805.98	1353.02
Dual-Process Theory	3761.01	1352.99
Enneagram of Personality Traits	3874.04	1348.96
Erikson’s Psychosocial Stages	3943.05	1430.95
Flow Theory	3873.00	1361.99
Freudian Psychoanalysis	3933.92	1355.08
Mental Models	3896.01	1367.99
Myers-Briggs Type Indicator	3903.03	1364.97
Social Cognitive Theory	3840.96	1339.04
User Design Persona	3972.96	1343.04
User-Centered Design	3864.89	1380.11

Table 7: Global Contingency Table of Final Decision Maker Successes and Failures by Framework.

The chi-squared test on the aggregated contingency table yielded a chi-squared statistic of 4.54 with 12 degrees of freedom and a p -value of 0.9715. This very high p -value suggests that any observed differences in the number of successes and failures across the description frameworks are entirely attributable to random variation.

Table 7 reports the aggregated contingency table for the Final Decision Maker, showing the total number of successes and failures for each description framework. As observed, the success and failure counts across the frameworks vary only slightly. Such minimal variation supports the conclusion that no single description framework offers a performance advantage in terms of Final Decision Maker accuracy. Thus, as in the case of the majority vote outcome, the choice of description framework, when evaluated based on

the final decision maker, should be guided by other factors.

2.3 Comparison Analysis: Final Decision Maker vs. Majority Vote

	Successes	Failures
Final Decision Maker	50398.81	17694.19
Majority Vote	50699.06	17393.94

Table 8: Global 2×2 Contingency Table.

We performed a global, weighted comparison of the Final Decision Maker method and the Majority Vote method to determine the most effective ensembling strategy. To this end, we aggregated the total number of successes and failures across all datasets. The overall accuracy for Final Decision Maker was 74.01%, while Majority Vote attained 74.46%. Although the Majority Vote shows a marginal advantage, the difference between the two methods is not statistically significant. We formally test this by stating the null hypothesis: H_0 : *There is no association between the ensembling method (Final Decision Maker vs. Majority Vote) and the success or failure outcome.* A chi-square test performed on the aggregated 2×2 contingency table (Table 8) yielded a chi-square statistic of 3.4609 with 1 degree of freedom and a p -value of 0.0628, leading us to fail to reject the null hypothesis.

Thus, we cannot conclude that one method is superior to the other overall.

3 Final Decision Maker and Majority Vote Finer Comparison

Weighted Mann–Whitney U test for pairwise comparison within datasets and the Weighted Wilcoxon Signed-Rank test for overall comparisons between the Final Decision Maker and Majority Vote approaches.

In this Section, we determine whether different description frameworks yield systematically distinct results to test the statistical differences between various description frameworks across datasets using non-parametric testing. Specifically, we compare the accuracy distributions of the Final Decision Maker and the Majority Vote ensembling methods. To achieve this, we employ the Weighted Mann–Whitney U test for pairwise comparisons [?] within

Dataset	Wilcoxon Statistic	p-value	Significant?
Strategy QA	0.0000	0.000000	Yes
Common Sense QA	9422141.5000	0.277116	No
Last Letters	125250.0000	0.000000	Yes
Social IQa	108766137.0000	0.000000	Yes
Social Support	17301336.0000	0.000000	Yes

Table 9: Weighted Wilcoxon Test Results: Final Decision Maker vs. Majority Vote.

datasets and the Weighted Wilcoxon Signed-Rank test [?] for overall comparisons between the Final Decision Maker and Majority Vote approaches. The Mann–Whitney U test is a powerful non-parametric statistical method employed to evaluate the null hypothesis. This hypothesis asserts that, for randomly selected values X and Y from two distinct populations, the probability of X being greater than Y is precisely equal to the probability of Y being greater than X . A comprehensive formulation establishes that observations from both groups are assumed to be independent, responses are at least ordinal to enable the comparison of any two observations to determine which is greater, and under the null hypothesis (H_0), the distributions of both populations are considered identical while the alternative hypothesis (H_1) asserts that the distributions are not identical. The Wilcoxon signed-rank test is a non-parametric statistical method used for hypothesis testing. For two matched samples, the Wilcoxon test functions as a paired difference test, similar to the paired Student’s t-test. This test is particularly useful when the assumption of normal distribution for differences is not met, as it only requires that the distribution of differences is symmetric around a central value. Its primary aim is to determine whether this central value significantly differs from zero.

The results of the Weighted Wilcoxon Signed-Rank test, summarized in Table 9, reveal whether the two decision paradigms (Final Decision Maker vs. Majority Vote) produce significantly different results for each dataset. In Strategy QA, Last Letter Concat, Social IQa, and Social Support the Final Decision Maker and Majority Vote approaches yield significantly different performance distributions. This implies that the choice of decision paradigm substantially affects model evaluation outcomes. No Significant Difference in Common Sense QA suggests that, for this dataset, either approach can be used interchangeably.

Given these findings, we proceed with pairwise comparisons of description frameworks within each dataset, focusing separately on the Final Decision Maker and Majority Vote approaches to determine whether certain frameworks consistently outperform others in specific dataset tasks.

3.1 Strategy QA Dataset

Weighted Mann–Whitney for Final Decision Maker, Dataset: strategyQA				
Framework 1	Framework 2	MW-stat	p-value	Sig?
Big Five Personality Traits	Cognitive Behavioral Theory	0.0000	0	Yes
Big Five Personality Traits	Cognitive Load Theory	0.0000	0	Yes
Big Five Personality Traits	Dual-Process Theory	0.0000	0	Yes
Big Five Personality Traits	Enneagram of Personality Traits	2485080.0000	0	Yes
Big Five Personality Traits	Erikson's Psychosocial Stages	0.0000	0	Yes
Big Five Personality Traits	Flow Theory	0.0000	0	Yes
Big Five Personality Traits	Freudian Psychoanalysis	0.0000	0	Yes
Big Five Personality Traits	Mental Models	0.0000	0	Yes
Big Five Personality Traits	Myers-Briggs Type Indicator	0.0000	0	Yes
Big Five Personality Traits	Social Cognitive Theory	2514720.0000	0	Yes
Big Five Personality Traits	User Design Persona	0.0000	0	Yes
Big Five Personality Traits	User-Centered Design	0.0000	0	Yes
Cognitive Behavioral Theory	Cognitive Load Theory	0.0000	0	Yes
Cognitive Behavioral Theory	Dual-Process Theory	2470508.0000	0	Yes
Cognitive Behavioral Theory	Enneagram of Personality Traits	2545614.0000	0	Yes
Cognitive Behavioral Theory	Erikson's Psychosocial Stages	0.0000	0	Yes
Cognitive Behavioral Theory	Flow Theory	0.0000	0	Yes
Cognitive Behavioral Theory	Freudian Psychoanalysis	0.0000	0	Yes
Cognitive Behavioral Theory	Mental Models	0.0000	0	Yes
Cognitive Behavioral Theory	Myers-Briggs Type Indicator	0.0000	0	Yes
Cognitive Behavioral Theory	Social Cognitive Theory	2575976.0000	0	Yes
Cognitive Behavioral Theory	User Design Persona	0.0000	0	Yes
Cognitive Behavioral Theory	User-Centered Design	0.0000	0	Yes
Cognitive Load Theory	Dual-Process Theory	2408608.0000	0	Yes
Cognitive Load Theory	Enneagram of Personality Traits	2481894.0000	0	Yes
Cognitive Load Theory	Erikson's Psychosocial Stages	2595628.0000	0	Yes
Cognitive Load Theory	Flow Theory	2553562.0000	0	Yes
Cognitive Load Theory	Freudian Psychoanalysis	2497474.0000	0	Yes
Cognitive Load Theory	Mental Models	2458524.0000	0	Yes
Cognitive Load Theory	Myers-Briggs Type Indicator	2494358.0000	0	Yes
Cognitive Load Theory	Social Cognitive Theory	2511496.0000	0	Yes
Cognitive Load Theory	User Design Persona	2541098.0000	0	Yes
Cognitive Load Theory	User-Centered Design	2537982.0000	0	Yes
Dual-Process Theory	Enneagram of Personality Traits	2462778.0000	0	Yes
Dual-Process Theory	Erikson's Psychosocial Stages	0.0000	0	Yes
Dual-Process Theory	Flow Theory	0.0000	0	Yes
Dual-Process Theory	Freudian Psychoanalysis	0.0000	0	Yes
Dual-Process Theory	Mental Models	0.0000	0	Yes
Dual-Process Theory	Myers-Briggs Type Indicator	0.0000	0	Yes
Dual-Process Theory	Social Cognitive Theory	2492152.0000	0	Yes
Dual-Process Theory	User Design Persona	0.0000	0	Yes
Dual-Process Theory	User-Centered Design	0.0000	0	Yes
Enneagram of Personality Traits	Erikson's Psychosocial Stages	0.0000	0	Yes
Enneagram of Personality Traits	Flow Theory	0.0000	0	Yes
Enneagram of Personality Traits	Freudian Psychoanalysis	0.0000	0	Yes
Enneagram of Personality Traits	Mental Models	0.0000	0	Yes
Enneagram of Personality Traits	Myers-Briggs Type Indicator	0.0000	0	Yes
Enneagram of Personality Traits	Social Cognitive Theory	2567916.0000	0	Yes
Enneagram of Personality Traits	User Design Persona	0.0000	0	Yes
Enneagram of Personality Traits	User-Centered Design	0.0000	0	Yes
Erikson's Psychosocial Stages	Flow Theory	2730574.0000	0	Yes
Erikson's Psychosocial Stages	Freudian Psychoanalysis	2670598.0000	0	Yes
Erikson's Psychosocial Stages	Mental Models	0.0000	0	Yes
Erikson's Psychosocial Stages	Myers-Briggs Type Indicator	2667266.0000	0	Yes
Erikson's Psychosocial Stages	Social Cognitive Theory	2685592.0000	0	Yes
Erikson's Psychosocial Stages	User Design Persona	2717246.0000	0	Yes
Erikson's Psychosocial Stages	User-Centered Design	2713914.0000	0	Yes
Flow Theory	Freudian Psychoanalysis	0.0000	0	Yes
Flow Theory	Mental Models	0.0000	0	Yes
Flow Theory	Myers-Briggs Type Indicator	2624039.0000	0	Yes
Flow Theory	Social Cognitive Theory	2642068.0000	0	Yes
Flow Theory	User Design Persona	2673209.0000	0	Yes
Flow Theory	User-Centered Design	2669931.0000	0	Yes
Freudian Psychoanalysis	Mental Models	0.0000	0	Yes
Freudian Psychoanalysis	Myers-Briggs Type Indicator	2566403.0000	0	Yes
Freudian Psychoanalysis	Social Cognitive Theory	2584036.0000	0	Yes
Freudian Psychoanalysis	User Design Persona	2614493.0000	0	Yes
Freudian Psychoanalysis	User-Centered Design	2611287.0000	0	Yes
Mental Models	Myers-Briggs Type Indicator	2526378.0000	0	Yes
Mental Models	Social Cognitive Theory	2543736.0000	0	Yes
Mental Models	User Design Persona	2573718.0000	0	Yes
Mental Models	User-Centered Design	2570562.0000	0	Yes
Myers-Briggs Type Indicator	Social Cognitive Theory	2580812.0000	0	Yes
Myers-Briggs Type Indicator	User Design Persona	2611231.0000	0	Yes
Myers-Briggs Type Indicator	User-Centered Design	2608029.0000	0	Yes
Social Cognitive Theory	User Design Persona	0.0000	0	Yes
Social Cognitive Theory	User-Centered Design	0.0000	0	Yes
User Design Persona	User-Centered Design	2656899.0000	0	Yes

Table 10: Pairwise Weighted Mann–Whitney comparisons of description frameworks for Strategy QA (Final Decision Maker).

The p-value analysis in 9 indicates a very strong statistical difference be-

tween the Final Decision Maker and Majority Vote approaches when weighing by the number of samples tested. Thus, we start the pair-wise analysis of the different description frameworks using the Final Decision Maker. According to the results, every comparison is statistically significant, meaning that here the choice of the description framework matters.

Weighted Mann-Whitney for Majority Vote, Dataset: strategyQA				
Framework 1	Framework 2	MW-stat	p-value	Sig?
Big Five Personality Traits	Cognitive Behavioral Theory	0.0000	0	Yes
Big Five Personality Traits	Cognitive Load Theory	0.0000	0	Yes
Big Five Personality Traits	Dual-Process Theory	2411760.0000	0	Yes
Big Five Personality Traits	Enneagram of Personality Traits	2485080.0000	0	Yes
Big Five Personality Traits	Erikson's Psychosocial Stages	0.0000	0	Yes
Big Five Personality Traits	Flow Theory	0.0000	0	Yes
Big Five Personality Traits	Freudian Psychoanalysis	0.0000	0	Yes
Big Five Personality Traits	Mental Models	0.0000	0	Yes
Big Five Personality Traits	Myers-Briggs Type Indicator	2497560.0000	0	Yes
Big Five Personality Traits	Social Cognitive Theory	2514720.0000	0	Yes
Big Five Personality Traits	User Design Persona	0.0000	0	Yes
Big Five Personality Traits	User-Centered Design	2541240.0000	0	Yes
Cognitive Behavioral Theory	Cognitive Load Theory	2489684.0000	0	Yes
Cognitive Behavioral Theory	Dual-Process Theory	2470508.0000	0	Yes
Cognitive Behavioral Theory	Enneagram of Personality Traits	2545614.0000	0	Yes
Cognitive Behavioral Theory	Erikson's Psychosocial Stages	0.0000	0	Yes
Cognitive Behavioral Theory	Flow Theory	0.0000	0	Yes
Cognitive Behavioral Theory	Freudian Psychoanalysis	0.0000	0	Yes
Cognitive Behavioral Theory	Mental Models	2521644.0000	0	Yes
Cognitive Behavioral Theory	Myers-Briggs Type Indicator	2558398.0000	0	Yes
Cognitive Behavioral Theory	Social Cognitive Theory	2575976.0000	0	Yes
Cognitive Behavioral Theory	User Design Persona	2606338.0000	0	Yes
Cognitive Behavioral Theory	User-Centered Design	2603142.0000	0	Yes
Cognitive Load Theory	Dual-Process Theory	2408668.0000	0	Yes
Cognitive Load Theory	Enneagram of Personality Traits	2481894.0000	0	Yes
Cognitive Load Theory	Erikson's Psychosocial Stages	0.0000	0	Yes
Cognitive Load Theory	Flow Theory	0.0000	0	Yes
Cognitive Load Theory	Freudian Psychoanalysis	0.0000	0	Yes
Cognitive Load Theory	Mental Models	2458524.0000	0	Yes
Cognitive Load Theory	Myers-Briggs Type Indicator	2494358.0000	0	Yes
Cognitive Load Theory	Social Cognitive Theory	2511496.0000	0	Yes
Cognitive Load Theory	User Design Persona	2541098.0000	0	Yes
Cognitive Load Theory	User-Centered Design	2537982.0000	0	Yes
Dual-Process Theory	Enneagram of Personality Traits	2462778.0000	0	Yes
Dual-Process Theory	Erikson's Psychosocial Stages	0.0000	0	Yes
Dual-Process Theory	Flow Theory	0.0000	0	Yes
Dual-Process Theory	Freudian Psychoanalysis	0.0000	0	Yes
Dual-Process Theory	Mental Models	0.0000	0	Yes
Dual-Process Theory	Myers-Briggs Type Indicator	0.0000	0	Yes
Dual-Process Theory	Social Cognitive Theory	0.0000	0	Yes
Dual-Process Theory	User Design Persona	0.0000	0	Yes
Dual-Process Theory	User-Centered Design	0.0000	0	Yes
Enneagram of Personality Traits	Erikson's Psychosocial Stages	0.0000	0	Yes
Enneagram of Personality Traits	Flow Theory	0.0000	0	Yes
Enneagram of Personality Traits	Freudian Psychoanalysis	0.0000	0	Yes
Enneagram of Personality Traits	Mental Models	0.0000	0	Yes
Enneagram of Personality Traits	Myers-Briggs Type Indicator	0.0000	0	Yes
Enneagram of Personality Traits	Social Cognitive Theory	0.0000	0	Yes
Enneagram of Personality Traits	User Design Persona	0.0000	0	Yes
Enneagram of Personality Traits	User-Centered Design	0.0000	0	Yes
Erikson's Psychosocial Stages	Flow Theory	2730574.0000	0	Yes
Erikson's Psychosocial Stages	Freudian Psychoanalysis	0.0000	0	Yes
Erikson's Psychosocial Stages	Mental Models	2628948.0000	0	Yes
Erikson's Psychosocial Stages	Myers-Briggs Type Indicator	2667266.0000	0	Yes
Erikson's Psychosocial Stages	Social Cognitive Theory	2685592.0000	0	Yes
Erikson's Psychosocial Stages	User Design Persona	2717246.0000	0	Yes
Erikson's Psychosocial Stages	User-Centered Design	2713914.0000	0	Yes
Flow Theory	Freudian Psychoanalysis	0.0000	0	Yes
Flow Theory	Mental Models	2586342.0000	0	Yes
Flow Theory	Myers-Briggs Type Indicator	2624039.0000	0	Yes
Flow Theory	Social Cognitive Theory	2642068.0000	0	Yes
Flow Theory	User Design Persona	2673209.0000	0	Yes
Flow Theory	User-Centered Design	2669931.0000	0	Yes
Freudian Psychoanalysis	Mental Models	2529534.0000	0	Yes
Freudian Psychoanalysis	Myers-Briggs Type Indicator	2566403.0000	0	Yes
Freudian Psychoanalysis	Social Cognitive Theory	2584036.0000	0	Yes
Freudian Psychoanalysis	User Design Persona	2614493.0000	0	Yes
Freudian Psychoanalysis	User-Centered Design	2611287.0000	0	Yes
Mental Models	Myers-Briggs Type Indicator	2536378.0000	0	Yes
Mental Models	Social Cognitive Theory	2543736.0000	0	Yes
Mental Models	User Design Persona	2573718.0000	0	Yes
Mental Models	User-Centered Design	2570562.0000	0	Yes
Myers-Briggs Type Indicator	Social Cognitive Theory	0.0000	0	Yes
Myers-Briggs Type Indicator	User Design Persona	0.0000	0	Yes
Myers-Briggs Type Indicator	User-Centered Design	0.0000	0	Yes
Social Cognitive Theory	User Design Persona	0.0000	0	Yes
Social Cognitive Theory	User-Centered Design	0.0000	0	Yes
User Design Persona	User-Centered Design	2656899.0000	0	Yes

Table 11: Pairwise Weighted Mann-Whitney comparisons of description frameworks for Strategy QA (Majority Vote).

We move on pair-wise analysis using the majority vote (Table 11). The table shows that nearly every pair comparison has a statistical significance, with p-values close to 0 for the vast majority of comparisons. As with the

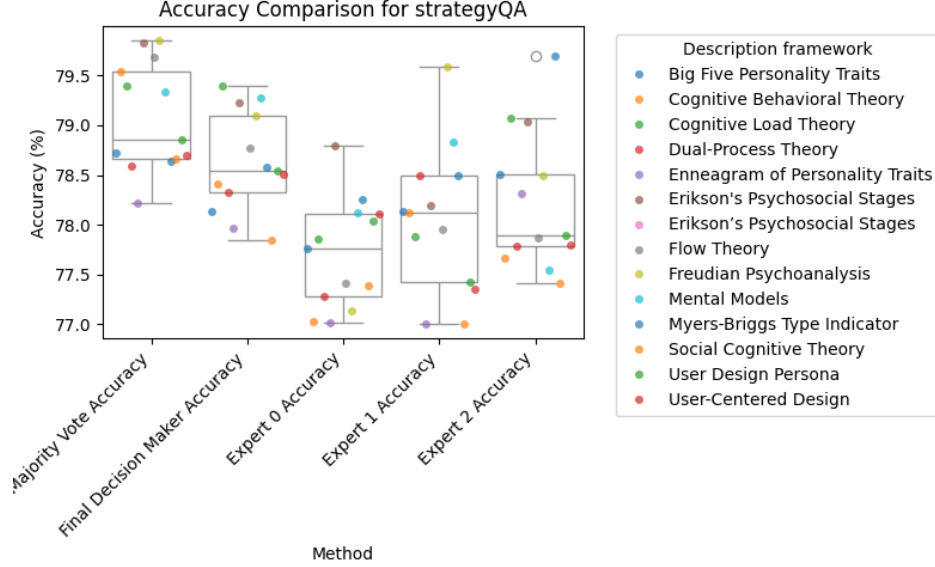


Figure 1: Box plot comparison Strategy QA dataset.

Final Decision Maker, each framework yields a unique accuracy distribution for the Majority Vote as well. The result is that description framework selection heavily impacts performance under the Majority Vote in this dataset, much as it does under the Final Decision Maker. In this dataset, both Final Decision Maker and Majority Vote results show that any two description frameworks significantly differ. Practically, if you are optimizing for performance on this dataset, you cannot assume frameworks are interchangeable. This finding is also supported by looking at the weighted mean differences shown in Figure 2. Also, as shown in Figure 1, the individual performance of the agents is typically lower than the adoption of an ensemble method.

3.2 Common Sense QA Dataset

The results reported in Table 9 revealed no statistical difference between choosing the Final Decision Maker and the Majority Vote ensembling method. Even though certain individual frameworks might show small differences, the overall weighted distribution of accuracy differences does not deviate enough to conclude that one approach is systematically superior, both methods perform comparably. But, comparing the description frameworks against each other, revealed consistent differences in every pair. Weighted Mann–Whitney

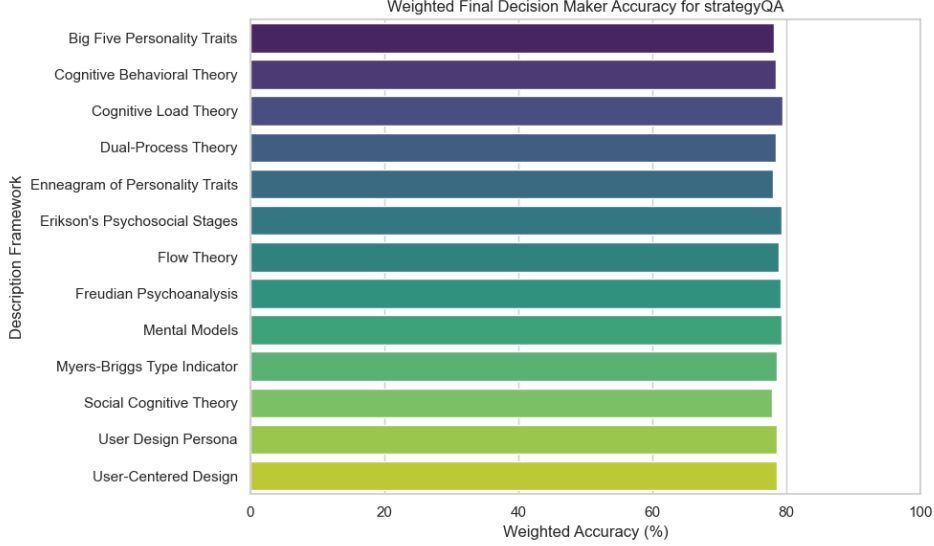


Figure 2: Weighted mean for the Strategy QA dataset.

test, reported in Table 13 shows that even small raw differences in final accuracy become significant. In practice, the results imply that each description framework yields a distinct Final Decision Maker accuracy distribution.

Much like the Final Decision Maker results, the statistical comparison reported in Table 12 reveals that all frameworks differ significantly under the Weighted Mann–Whitney for the Majority Vote approach as well.

These findings are also supported by analyzing the weighted mean accuracies shown in Figure 4. Indeed, a graphical representation of the results obtained in this dataset is proposed in Figure 3. Adopting an ensemble method generally provides better performance w.r.t. the single expert agents, such as Big Five or Cognitive Load Theory.

3.3 Last Letter Concat Dataset

The analysis results reported in Table 9 revealed a p-value near zero for this dataset, indicating a clear difference between the Final Decision Maker and Majority Vote. Indeed, the Weighted Wilcoxon test strongly rejects the null hypothesis that the two methods produce the same distribution of accuracies. Both ensembling approaches achieve overall high accuracies (often above 90%), making them particularly interesting since the difference is consistent and robust even within a high-performing environment.

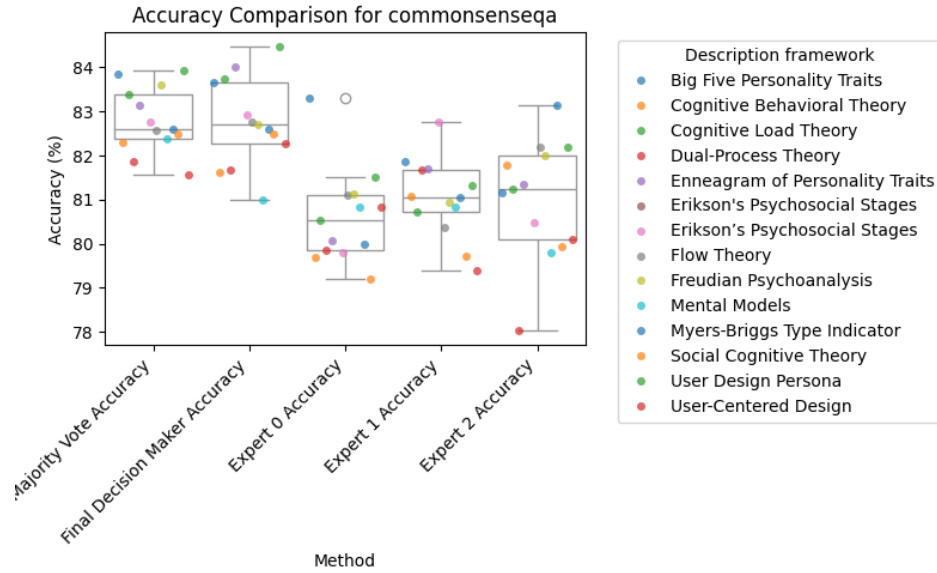


Figure 3: Box plot comparison Common Sense QA dataset.

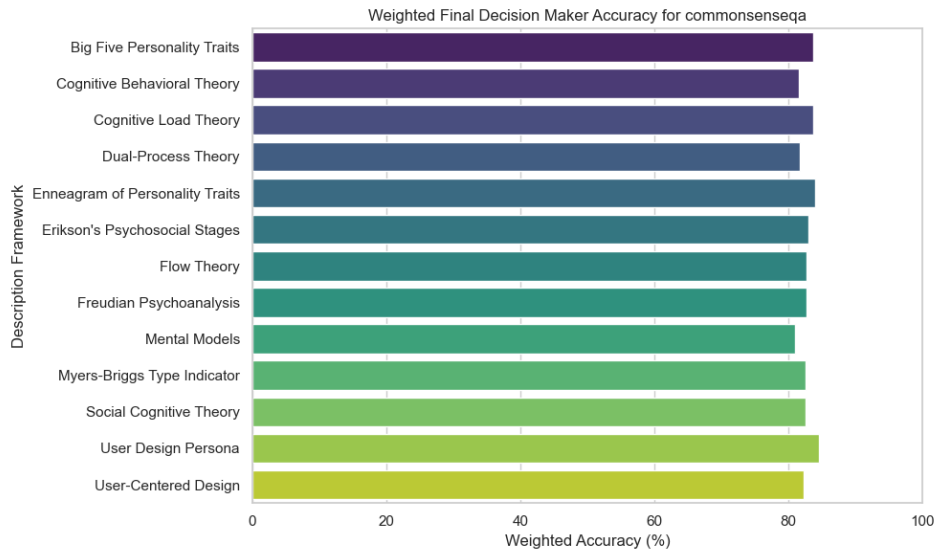


Figure 4: Weighted mean for the Common Sense QA dataset.

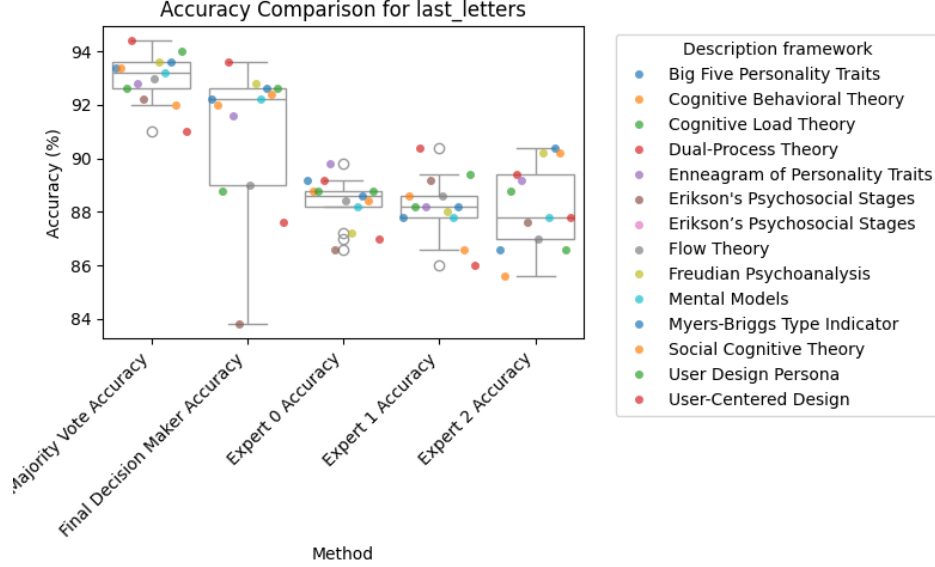


Figure 5: Boxplot Last Letter Concat dataset.

Starting a deeper analysis of the Final Decision Maker, proposed in Table 14, we see that almost all pairs comparisons are statistically significant, with only a few exceptions (for example, Big Five Personality Traits and Mental Models, Myers-Briggs Type Indicator and User Design Persona). This indicates that certain frameworks effectively yield the same final accuracy distribution for the Final Decision Maker. The large majority of frameworks stand out from each other, but those scattered no statistically significant results show that for certain frameworks, the Weighted Mann–Whitney test sees no meaningful difference. Analyzing the statistically significant comparisons for the Majority Vote (Table 15) we note the same phenomenon. Indeed, almost every comparison is significant, but again, there are a small number of exceptions. For instance, Big Five Personality Traits and Cognitive Behavioral Theory, Freudian Psychoanalysis and Myers-Briggs Type Indicator are not significant. The overall pattern remains that most pairs differ significantly, implying that in Last Letters, the chosen framework heavily influences final accuracy with the Majority Vote approach. Interestingly, in this dataset, our approach typically has very high accuracies (90%+). Still, small differences are statistically significant.

The statistical findings are also supported by the analysis of the weighted mean accuracy plot shown in Figure 6. Furthermore, a visual analysis of

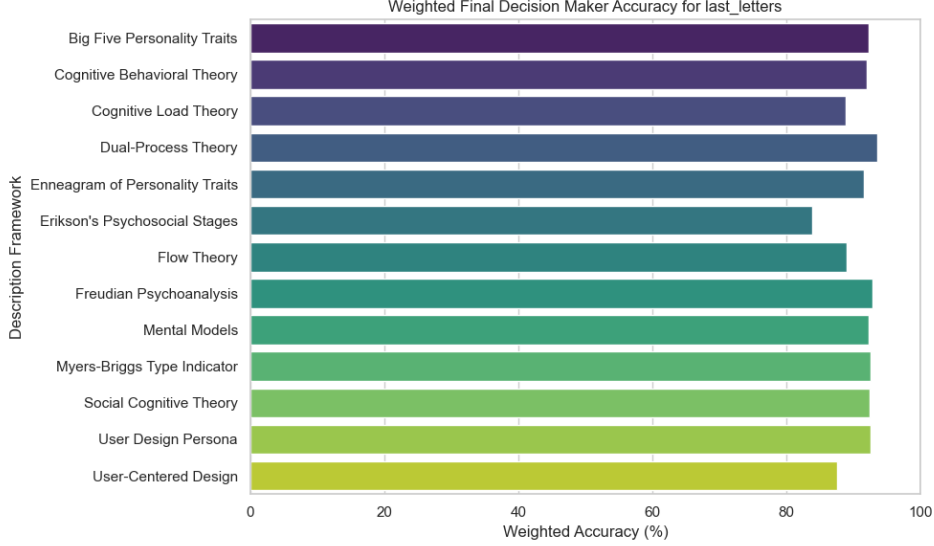


Figure 6: Weighted mean for the Last Letter Concat dataset.

the performance distribution presented in Figure 5 indicates a significant variability in scores for the Final Decision Maker, whereas the scores for the Majority Vote demonstrate a more compact distribution. Furthermore, the performance levels of the individual agents are considerably lower when compared to the two ensemble methods.

3.4 Social IQa Dataset

The results of the Weighted Wilcoxon test, shown in Table 9, indicate a highly significant difference between the Final Decision Maker and Majority Vote methods in this dataset, with a p-value effectively close to 0. This suggests that, from a weighted perspective, one of these methods consistently outperforms the other for the SocialIQa dataset. The findings strongly imply that the two methods are not interchangeable regarding their final accuracy outcomes.

The statistical analysis of the Final Decision Maker presented in Table 16 shows that all or nearly all pairs of frameworks are statistically relevant. For instance, both Dual-Process Theory and Flow Theory exhibit significant differences. This means that every pair of description frameworks demonstrates a statistically significant difference when evaluated using the Weighted Mann–Whitney test. As a result, even small raw differences can

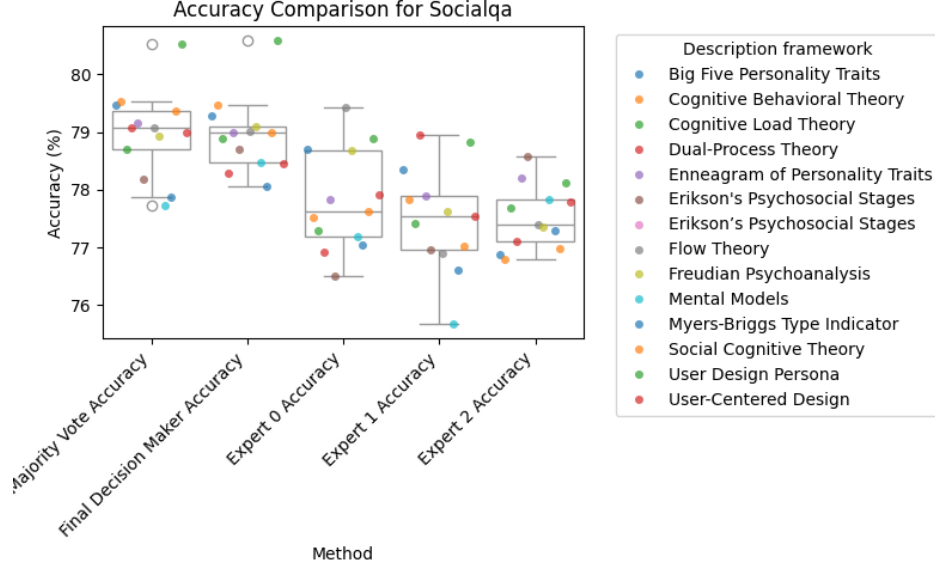


Figure 7: Boxplot Social IQa dataset.

yield highly significant outcomes. Practically speaking, this suggests that we cannot treat any two descriptive frameworks as equivalent—each framework produces a distinct accuracy distribution for the Final Decision Maker from the perspective of the Weighted Mann–Whitney analysis.

The Majority Vote analysis, reported in Table 17, shows an extensive series of statistically valid comparisons, with very few exceptions (e.g. Dual-Process Theory and Flow Theory). As with Final Decision Maker, Majority Vote analysis demonstrates that almost all frameworks differ in Weighted Mann–Whitney.

Also, the weighted mean accuracy presented in Figure 8 supports the findings. Furthermore, the distribution of performance scores illustrated in Figure 7 highlights two significant observations. Firstly, the Majority Vote ensembling method demonstrates a greater degree of variability in performance compared to the Final Decision Maker. Secondly, the Cognitive Load Theory description framework emerges as the highest scorer, whereas the Myers-Briggs Type Indicator description framework consistently ranks as the lowest performer across all assessments.

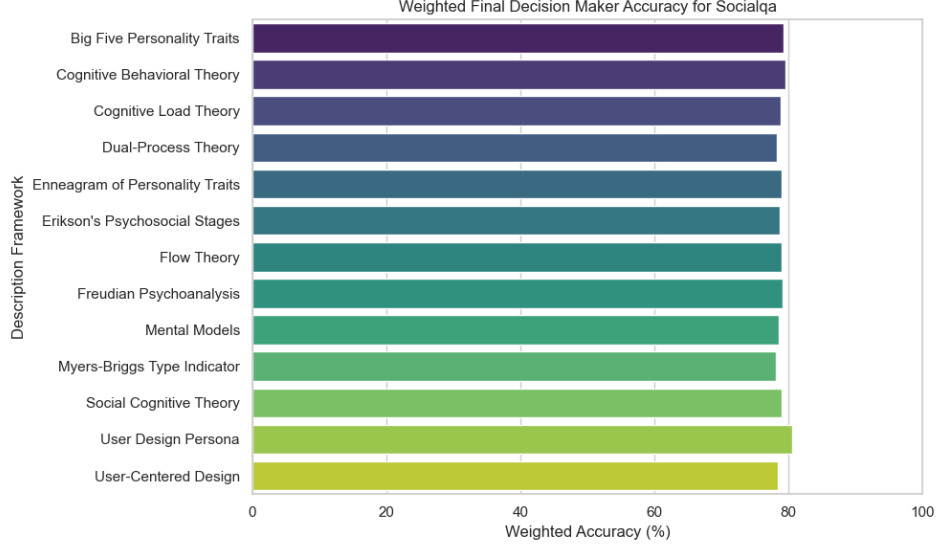


Figure 8: Weighted mean for the Social IQa dataset.

3.5 Social Support Dataset

In this dataset, all the description frameworks have fairly low overall accuracies (in the 40% range). The Weighted Wilcoxon test results presented in Table 9 yield a p-value near zero, implying a significant difference between the two methods once weighting (based on the number of samples tested) is accounted for. Even though the absolute performance might be low, the difference is systematically observed across the various description frameworks. In practice, this indicates that, in this dataset, one approach has a measurable advantage.

The analysis of the Final Decision Maker proposed in Table 18, demonstrates many statistically valid comparisons, with few exceptions. For instance, Big Five Personality Traits and Cognitive Load Theory are not statistically different. Although many comparisons are significant, certain frameworks produce statistically indistinguishable Weighted Mann–Whitney outcomes. Considering this dataset tends to have low accuracies (in the 40% range), the significance indicates that even small improvements are consistent. The large number of statistically valid comparisons confirms that the choice of framework typically matters under the Final Decision Maker approach. The comparative analysis of the Majority Vote (reported in Table 19) shows a similar trend of statistically significant comparisons for the Final

Decision Maker (with few exceptions such as Cognitive Behavioral Theory and Enneagram of Personality Traits, or Cognitive Behavioral Theory and Flow Theory).

The permutation test (20) is particularly suitable for comparing macro F1-scores due to its non-parametric nature and flexibility in handling complex, aggregated metrics. Unlike parametric tests, permutation tests do not assume a specific distribution of the data, making them ideal for evaluating metrics like the macro F1-score, which is the unweighted average of F1-scores across all classes in a multi-class classification problem. In practice, the permutation test involves repeatedly shuffling the labels of the dataset and recalculating the performance metric to generate a distribution of scores under the null hypothesis—that there is no difference between the models. By comparing the observed difference in macro F1-scores to this null distribution, we can determine the empirical p-value, indicating the likelihood of observing such a difference by chance. This method is particularly advantageous when dealing with metrics like the macro F1-score, which can be sensitive to class imbalances and may not meet the assumptions required for parametric tests. By not relying on such assumptions, the permutation test provides a more reliable assessment of statistical significance in these contexts. Therefore, the permutation test is a robust and appropriate choice for comparing macro F1-scores, offering a non-parametric approach that accommodates the complexities inherent in multi-class classification evaluations.

Finally, the analysis of the performance score distributions presented in Figure 9 indicates that cognitive theory-based description frameworks consistently yield superior results. This finding underscores the importance of cognitive skills for agents in effectively addressing the task at hand. The same finding is supported by the visual analysis of the weighted mean of accuracy presented in Figure 10.

4 Agent Roles Analyses

We first tested whether expert roles differ statistically in their accuracy scores within each dataset by applying weighted analyses that account for the number of samples tested per framework. Specifically, we used a weighted ANOVA (also known as Welch’s ANOVA)[?] to compare the weighted means and a weighted Kruskal–Wallis test. We chose Welch’s t-test, or unequal variances t-test since it is a two-sample test to test the (null) hypothesis that two populations have equal means. This statistic is more reliable when the two samples have unequal variances and possibly unequal sample sizes. This

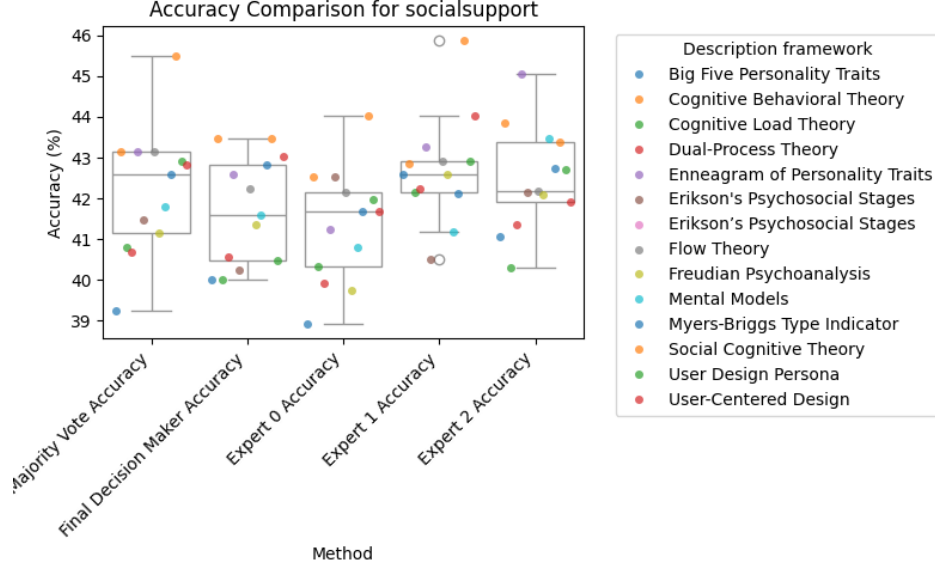


Figure 9: Boxplot Social Support dataset.

is the right choice in our cases since we cannot assume that the distributions analyzed have equal variances and they do not have equal sample sizes. Furthermore, we make the ANOVA findings more robust by also applying the Kruskal–Wallis test. The Kruskal–Wallis test [?], or one-way ANOVA on ranks, is a non-parametric statistical test to test whether samples originate from the same distribution. However, a significant test indicates that at least one sample stochastically dominates one other sample without identifying where this stochastic dominance occurs or for how many pairs of groups of stochastic dominance obtains. Thus, when the weighted ANOVA yielded a significant overall effect ($p < 0.05$), we conducted Tukey’s HSD post-hoc pairwise comparisons [?] to determine which specific roles differ. Turkey’s HSD post-hoc pairwise test is a single-step multiple comparison procedure and statistical test used to correctly interpret the statistical significance of the difference between means. The test, based on a studentized range distribution, compares all possible pairs of means in a distribution similar to the distribution of t from the t -test. The test compares the means of every treatment to the means of every other treatment; that is, it applies simultaneously to the set of all pairwise comparisons.

The overall weighted ANOVA and Kruskal–Wallis results are summarized in Table 21 and Table 22, respectively, while the detailed Tukey’s HSD results

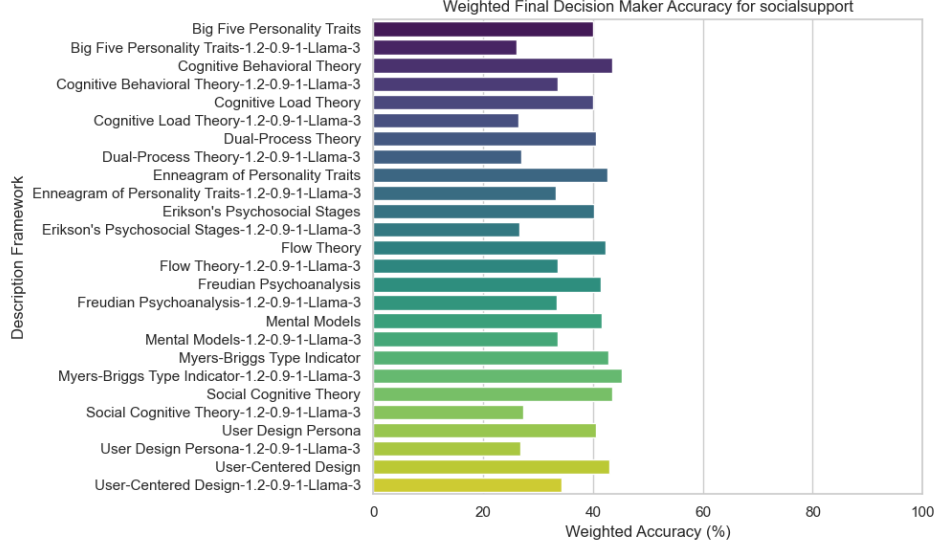


Figure 10: Weighted mean for the Social Support dataset.

are reported in Tables 24, 25, and 26. The distribution of expert roles within each dataset is presented in Table 23. Here, we discuss each dataset in isolation.

4.1 Strategy QA Dataset

The weighted ANOVA analysis reported in Table 21 indicates a statistically significant difference in the weighted mean accuracy among expert roles, while the weighted Kruskal–Wallis test (Table 22) did not detect differences. But, since the results of the multiple comparisons may inflate the Type I error rate ¹ we proceed with deeper analyses.

Analyzing the distribution of roles, shown in Figure 11, we noted that the most adopted role is Artificial Intelligence, followed by Cognitive Psychology, and Cognitive Science. This indicates that the Project Manager Agent thinks that Cognitive skills are important and so required to efficiently perform the task.

We performed the Tukey’s HSD pairwise comparisons (Table 24) and we discover that :

¹Type I error rate is the probability of incorrectly rejecting a true null hypothesis (H_0). It is the likelihood of concluding that there is an effect or difference when none actually exists. This is often referred to as a *false positive* result.



Figure 11: Expert roles distribution for the Strategy QA dataset.

- *Artificial Intelligence* significantly differs from *Cognitive Psychology*, *Logic and Formal Reasoning*, *Machine Learning*, and *Natural Language Processing*.
- *Cognitive Psychology* is significantly different from *Mathematical Logic*.
- *Cognitive Science* is significantly different from both *Logic and Formal Reasoning* and *Machine Learning*.
- *Formal Logic* differs significantly from *Logic and Formal Reasoning* and *Machine Learning*.
- *Machine Learning* is significantly different from *Mathematical Logic*, and *Mathematical Logic* differs significantly from *Natural Language Processing*.

In this dataset, the weighted ANOVA indicates that expert role selection significantly affects accuracy. The distribution of roles shown in Figure 11 together with Tukey’s HSD test reveal that roles selection has an impact on performance. Indeed *Artificial Intelligence*, *Machine Learning*, *Logic and Formal Reasoning*, and *Mathematical Logic* exhibit statistically significant differences.

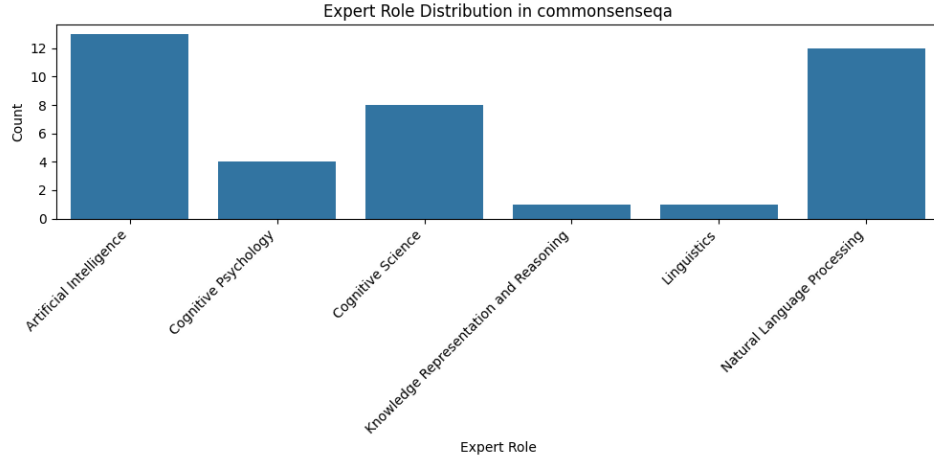


Figure 12: Expert roles distribution for the Common Sense QA dataset.

4.2 Common Sense QA Dataset

The analysis of the weighted ANOVA test and the Kruskal–Wallis test, reported respectively in Table 21 and Table 22, revealed very high p-values, indicating that role selection does not have a decisive impact on accuracy. Although the role distribution shown in Figure 12 exhibits variability, with a strong preference for *Artificial Intelligence* and *Natural Language Processing* roles, the differences in the selection are not statistically significant.

4.3 Last Letter Concat Dataset

The statistical results of the weighted ANOVA test and the Kruskal–Wallis test, reported respectively in Table 21 and Table 22, did not indicate a statistical valid difference in role selection. Similar to the Common Sense QA dataset.

The role distribution, shown in Figure 13, we discover a preference for *Computational Linguistics*, *Morphology*, *Phonetics* and *Cryptography* roles. However, this selection is not statistically valid. The uniform sample size and the non-significant test results imply that expert role selection does not affect accuracy in this dataset.

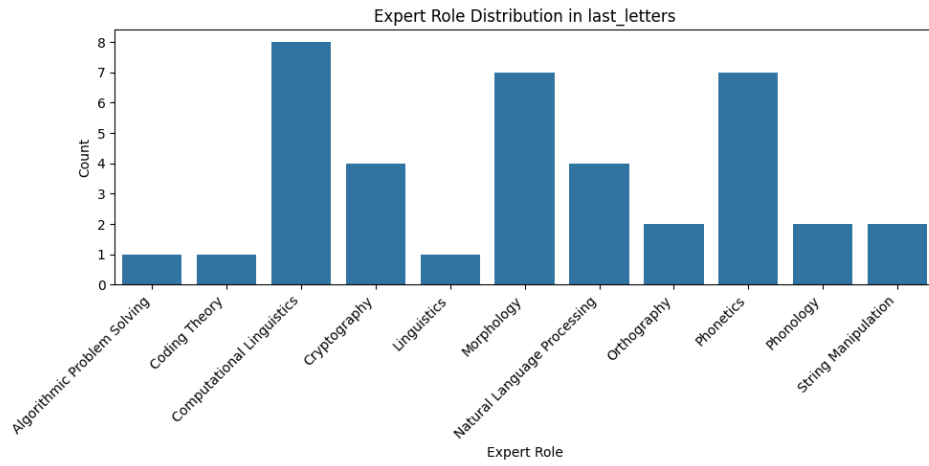


Figure 13: Expert roles distribution for the Last Letter Concat dataset.

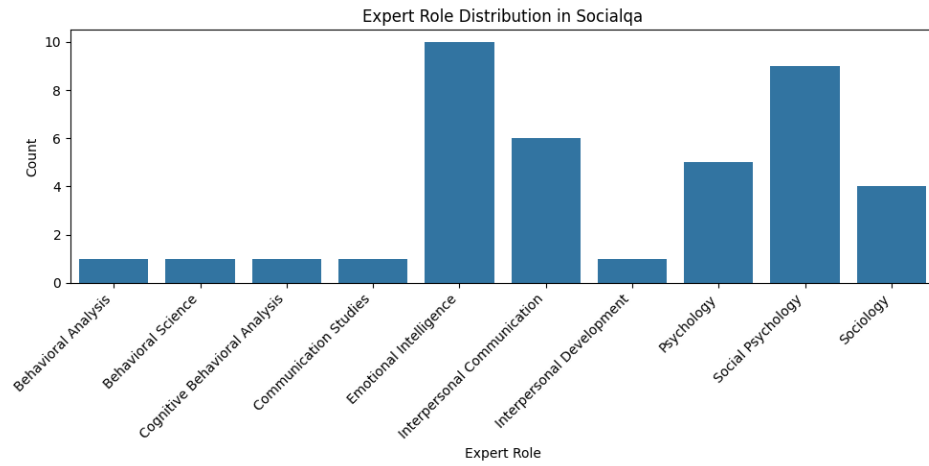


Figure 14: Expert roles distribution for the Social IQa dataset.

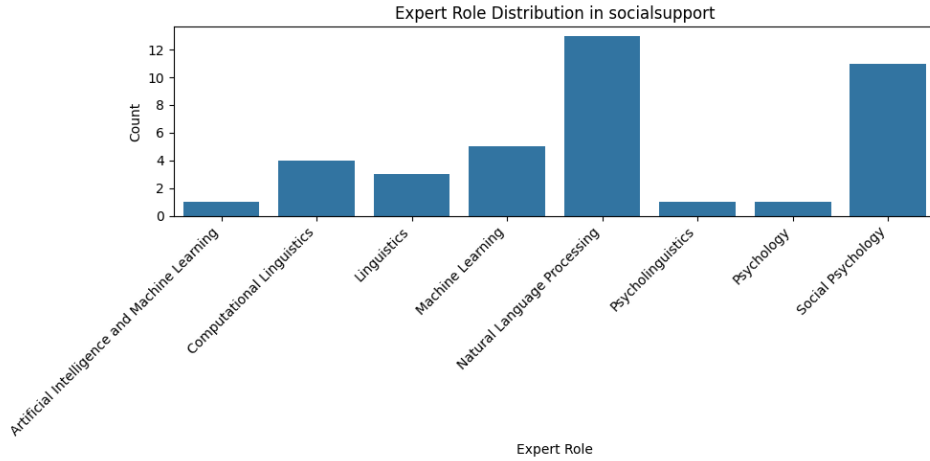


Figure 15: Expert roles distribution for the Social Support dataset.

4.4 Social IQa Dataset

The statistical test results reported in Table 21 and Table 22 revealed that while the weighted ANOVA is highly significant, the weighted Kruskal–Wallis test is marginal. We further analyzed the possibility of a role selection effect by conducting Tukey’s HSD analysis. The test results, reported in Table 25, indicate that only the *Behavioral Analysis* role is significantly different from *Behavioral Science*, *Cognitive Behavioral Analysis*, *Communication Studies*, *Emotional Intelligence*, *Interpersonal Communication*, *Interpersonal Development*, *Psychology*, *Social Psychology*, and *Sociology*.

We further analyzed the role distribution, as shown in Figure 14. The distribution demonstrates that the set of roles comprises strong task-dependent roles, including *Emotional Intelligence*, *Interpersonal Communication*, *Psychology*, *Social Psychology*, and *Sociology*.

Overall, the significant ANOVA and detailed Tukey’s HSD pairwise comparisons indicate that, in the Social IQa dataset, expert role selection plays an important role. In particular, the fact that *Behavioral Analysis* is statistically distinct from several other roles suggests that its inclusion (or exclusion) could meaningfully affect system accuracy.

4.5 Social Support Dataset

The statistical tests (reported in Table 21 and Table 22) both indicate statistically significant values. We therefore proceed with a finer comparison

using Tukey’s HSD pairwise test. The results, reported in Table 26, identify only one significant difference—between *Natural Language Processing* and *Social Psychology*.

We also analyzed the role distribution shown in Figure 15, which reveals a strong preference for both *Natural Language Processing* and *Social Psychology* roles. To sum up, while the overall tests indicate that expert role selection affects accuracy, the limited number of significant pairwise differences suggests that the variations among roles are relatively modest.

4.6 Overall Analysis

Across the examined datasets, our analyses reveal a strong impact of expert role selection on task accuracy. In the Common Sense QA and Last Letter Concat datasets (see Sections 4.2 and 4.3), the weighted ANOVA and Kruskal–Wallis tests did not show statistically significant differences in role selection. However, an inspection of the role distributions indicates that each dataset exhibits distinct preferred roles. This pattern suggests that the Project Manager agent is able to select the appropriate expertise based on the specific task requirements, resulting in strong task-dependent role allocation despite overall non-significant differences.

In contrast, the analysis of the Social IQa dataset (subsection 4.4), Strategy QA dataset (subsection 4.1), and Social Support dataset (subsection 4.5) indicate that expert role selection plays an important role. The weighted ANOVA yielded highly significant results, and Tukey’s HSD pairwise comparisons reveal statistically valid role selection. This implies that the inclusion (or exclusion) of some roles can meaningfully affect system accuracy in these tasks.

Overall, these findings demonstrate that the influence of expert role selection is highly task-dependent. In some datasets, such as Common Sense QA and Last Letter Concat, the impact is relatively modest, whereas in others, like Social IQa, role selection plays a critical role in shaping performance. This underscores not only the importance of task-dependent role allocation strategies in multi-agent systems, but also the efficacy of the Project Manager agent in selecting the most important role for solving a text. The Project Manager agent exhibits the flexibility necessary to successfully select the appropriate roles for the expert agents to solve tasks.

5 Agent Position Bias

We investigated whether the position of an agent (i.e., Expert 0, Expert 1, or Expert 2) influences performance. To conduct this evaluation, we aggregated the accuracy data from all datasets by weighting each observation by the number of samples tested. The aggregated counts yield the contingency table presented in Table 27. A chi-square test was then performed on this 3×2 contingency table. The null hypothesis H_0 states that: *the expert’s position does not affect performance*, i.e., all three positions yield the same success-to-failure ratio. The test produced a chi-square statistic of $\chi^2 = 2.28$ with 2 degrees of freedom and a p -value of approximately 0.32. Since the p -value is greater than 0.05 we fail to reject the null hypothesis.

These results indicate that there is no statistically significant positional bias in the performance of the agents. Although minor numerical differences exist among the aggregated successes of Experts 0, 1, and 2, these differences are well within the bounds of random variation given the sample sizes. Consequently, we conclude that the agents perform equivalently regardless of their position.

6 Task Instructions and Contexts

In this section, we report the *task instructions* and the *context* for each dataset-task.

Note: We use the same task instructions and context for all the description frameworks of the same dataset.

As *context*, we generally use the one provided in the dataset, downloaded from https://github.com/google/BIG-bench/tree/main/bigbench/benchmark_tasks, if available.

6.1 Common Sense QA Dataset

task-instruction: Answer closed-ended questions by indicating the capital letter of the correct answer from the five alternatives provided

context: commonsense question answering

example of query: Question: "Where would you find magazines along side many other printed works?" A) doctor B) bookstore C) market D) train station E) mortuary

6.2 Last Letter Concat Dataset

task-instruction: Take the last letters of the words and concatenate them

context: linguistics puzzles

example of query: Take the last letters of each words in "Silvia Carolina Stan Chuck" and concatenate them.

6.3 Social IQa Dataset

task-instruction: Answer to multi choice questions

context: Questions are about the motivations, emotional reactions, and preceding and following events surrounding interpersonal situations

example of query: Answer the following question by reporting the capital letter of the correct answer among the alternatives: ""Tracy didn't go home that evening and resisted Riley's attacks. What does Tracy need to do before this? "" A) Make a new plan B) Go home and see Riley C) Find somewhere to go

6.4 Strategy QA Dataset

task-instruction: Answer multiple-choice questions where the required reasoning steps are implicit in the question

context: The task aims to measure the ability of pre-trained models on context-free question answering, multi-step, implicit reasoning, and logical reasoning

example of query: Answer the following question by reporting the capital letter of the correct answer among the alternatives: ""Is it common to see frost during some college commencements?"" A) Yes B) No

6.5 Social Support Dataset

task-instruction: Classify whether a reply is unsupportive, neutral, or supportive

context: The task aims to measure the ability of pre-trained models on understanding supportive (and unsupportive) language uses for social support classification

example of query: Answer the following question by reporting the capital letter of the correct answer among the alternatives: ""While I don't have any source offhand, there are authentic Jewish sources that mentions the idea that 7 represents nature (7 days of creation) and 8 is supernatural. It

has nothing to do with the shape of the number.""" Q: Is the following reply unsupportive, neutral, or supportive? A) unsupportive B) neutral C) supportive

7 Implementation Details

We implemented our approach by adopting the Meta Llama 3.1-70B Instruct large language model (LLM). The inference process is guided by a temperature setting of 1.2, which introduces a moderate degree of randomness in the model’s responses, allowing for more diverse and creative outputs. Additionally, nucleus sampling is set at 0.9, ensuring that the model considers only the most relevant and probable tokens while generating text, striking a balance between diversity and coherence. Finally, in this first set of experiments, we do not focus on obtaining the highest probability answer from the LLM. Instead, we configure the system to generate a single alternative response for each input, ensuring a deterministic output. We ensure reproducibility by setting a fixed seed value for random number generation directly within our implementation files.

Additionally, we leverage the default *chat-template* provided by Llama 3.1 to structure interactions between the system and the user. This template explicitly defines the roles of both the system and the user, ensuring that the model correctly interprets input prompts and maintains consistency in its responses. By adhering to this standardized template, we enhance the model’s ability to generate contextually appropriate and structured outputs, improving overall response coherence and alignment with the intended person assigned to the agent.

7.1 Computational Cost Estimation

We conduct our experiments using the CINECA HPC infrastructure. This section provides an estimation of the required GPU hours.

For each of the five datasets, we run 13 description frameworks. Each experiment is executed on 3 H100 GPUs. Typically, processing 500 experimental samples requires approximately 24 hours. Below, we report the number of samples tested for each dataset and the corresponding estimated computational cost.

- **Strategy QA:** More than 1600 samples, but we consider 1500 for computational cost estimation. Since we process 500 samples per day, the experiments require 3 days. The computational cost is:

$$GPU\ hours\ cost = 3 \times 13 \times 24 \times 3 = 2808$$

- **Common Sense QA:** Approximately 500 samples. As we process 500 samples per day, the experiments require 1 day. The computational cost is:

$$GPU\ hours\ cost = 3 \times 13 \times 24 \times 1 = 936$$

- **Last Letter Concat:** 500 samples. Since we process 500 samples per day, the experiments require 1 day. The computational cost is:

$$GPU\ hours\ cost = 3 \times 13 \times 24 \times 1 = 936$$

- **Social IQa:** More than 1700 samples, but we consider 1500 for computational cost estimation. Since we process 500 samples per day, the experiments require 3 days. The computational cost is:

$$GPU\ hours\ cost = 3 \times 13 \times 24 \times 3 = 2808$$

- **Social Support:** 897 samples, rounded to 1000 for computational cost estimation. Since we process 500 samples per day, the experiments require 2 days. The computational cost is:

$$GPU\ hours\ cost = 3 \times 13 \times 24 \times 2 = 1872$$

Thus, the total estimated computational cost is:

$$Total\ GPU\ hours\ cost = 2808 + 936 + 936 + 2808 + 1872 = 9360$$

7.2 System Usage

Each parameter in the JSON file plays a specific role in guiding the system’s behavior. Below we report a detailed explanation of each parameter that must be included in the JSON configuration file.

- **name:** The name of the experiment.
- **task:** the task instruction to provide to the experts.
- **context:** The broader domain of the task, indicating that it belongs to, e.g., commonsense question answering.

- **description_framework**: The theoretical description framework guiding the generation of the agent description profiles.
- **model_name**: The specific language model used, in this case, Meta Llama 3.1-70B Instruct.
- **output_dir**: The directory where the generated results are stored.
- **input**: The input file containing queries, with each query on a separate line.
- **temperature**: Controls response randomness. We adopt a value of 1.2 to introduce moderate variability, allowing for diverse outputs.
- **nucleus**: Defines the nucleus sampling threshold. We adopt a value of 0.9 ensures that the model considers only the most relevant tokens during generation.
- **alternatives**: The number of alternative outputs generated per query. Here, it is set to 1 to ensure a deterministic response.
- **resume**: A flag indicating whether to resume execution from a previous run.
- **cache_dir**: Specifies the directory for caching model weights and computations. This parameter avoids downloading the same model multiple times.
- **max_experts_number**: The maximum number of expert models used during inference. In our experiments, we fix this parameter to 3.
- **token**: The authentication token required to access the model on the HuggingFace repository.

This configuration setup allows users to customize the system’s behavior while ensuring reproducibility across different experimental settings.

To execute our system, we utilize a SLURM job script to manage resource allocation and scheduling efficiently. Below, we provide an example SLURM configuration file:

```
#!/bin/bash
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=16
#SBATCH --gpus=3
```

```
#SBATCH --mem=100GB
#SBATCH --time="23:59:59"
#SBATCH --partition="cluster-partition"
#SBATCH --qos=normal
#SBATCH --job-name="job-name"
#SBATCH --output="std out"
#SBATCH --error="std err"

echo "Job starting"

source /myvenv/bin/activate

python ../PoE_Small/config_file.py
--config-file configuration.json
```

After specifying the SLURM directives, the script activates a virtual environment ('source /myvenv/bin/activate') and executes the system using a Python script with a predefined JSON configuration file. This setup ensures efficient resource utilization and facilitates job execution on a high-performance computing cluster (e.g., CINECA pre-exascale HPC).

Weighted Mann–Whitney for Majority Vote, Dataset: Common Sense QA				
Framework 1	Framework 2	MW-stat	p-value	Sig?
Big Five Personality Traits	Cognitive Behavioral Theory	318047.0000	4.48945e-247	Yes
Big Five Personality Traits	Cognitive Load Theory	311920.0000	1.10396e-244	Yes
Big Five Personality Traits	Dual-Process Theory	304122.0000	1.21835e-241	Yes
Big Five Personality Traits	Enneagram of Personality Traits	313591.0000	2.45995e-245	Yes
Big Five Personality Traits	Erikson's Psychosocial Stages	319718.0000	1.00039e-247	Yes
Big Five Personality Traits	Flow Theory	303565.0000	2.00063e-241	Yes
Big Five Personality Traits	Freudian Psychoanalysis	315819.0000	3.32321e-246	Yes
Big Five Personality Traits	Mental Models	322503.0000	8.19354e-249	Yes
Big Five Personality Traits	Myers-Briggs Type Indicator	320275.0000	6.065e-248	Yes
Big Five Personality Traits	Social Cognitive Theory	305236.0000	4.47797e-242	Yes
Big Five Personality Traits	User Design Persona	0.0000	1.65011e-247	Yes
Big Five Personality Traits	User-Centered Design	308021.0000	3.6674e-243	Yes
Cognitive Behavioral Theory	Cognitive Load Theory	0.0000	1.00039e-247	Yes
Cognitive Behavioral Theory	Dual-Process Theory	311766.0000	1.10396e-244	Yes
Cognitive Behavioral Theory	Enneagram of Personality Traits	0.0000	2.22921e-248	Yes
Cognitive Behavioral Theory	Erikson's Psychosocial Stages	0.0000	9.06613e-251	Yes
Cognitive Behavioral Theory	Flow Theory	0.0000	1.82095e-244	Yes
Cognitive Behavioral Theory	Freudian Psychoanalysis	0.0000	3.01157e-249	Yes
Cognitive Behavioral Theory	Mental Models	0.0000	7.42564e-252	Yes
Cognitive Behavioral Theory	Myers-Briggs Type Indicator	0.0000	5.49647e-251	Yes
Cognitive Behavioral Theory	Social Cognitive Theory	0.0000	4.0576e-245	Yes
Cognitive Behavioral Theory	User Design Persona	0.0000	1.49541e-250	Yes
Cognitive Behavioral Theory	User-Centered Design	315763.0000	3.32322e-246	Yes
Cognitive Load Theory	Dual-Process Theory	305760.0000	2.71479e-242	Yes
Cognitive Load Theory	Enneagram of Personality Traits	315280.0000	5.48151e-246	Yes
Cognitive Load Theory	Erikson's Psychosocial Stages	321440.0000	2.22921e-248	Yes
Cognitive Load Theory	Flow Theory	305200.0000	4.47797e-242	Yes
Cognitive Load Theory	Freudian Psychoanalysis	0.0000	7.40516e-247	Yes
Cognitive Load Theory	Mental Models	324240.0000	1.8258e-249	Yes
Cognitive Load Theory	Myers-Briggs Type Indicator	322000.0000	1.35148e-248	Yes
Cognitive Load Theory	Social Cognitive Theory	306880.0000	9.97808e-243	Yes
Cognitive Load Theory	User Design Persona	0.0000	3.67698e-248	Yes
Cognitive Load Theory	User-Centered Design	309680.0000	8.17197e-244	Yes
Dual-Process Theory	Enneagram of Personality Traits	0.0000	6.04927e-243	Yes
Dual-Process Theory	Erikson's Psychosocial Stages	0.0000	2.45996e-245	Yes
Dual-Process Theory	Flow Theory	0.0000	4.9423e-239	Yes
Dual-Process Theory	Freudian Psychoanalysis	0.0000	8.17198e-244	Yes
Dual-Process Theory	Mental Models	0.0000	2.01474e-246	Yes
Dual-Process Theory	Myers-Briggs Type Indicator	0.0000	1.49137e-245	Yes
Dual-Process Theory	Social Cognitive Theory	0.0000	1.10125e-239	Yes
Dual-Process Theory	User Design Persona	0.0000	4.0576e-245	Yes
Dual-Process Theory	User-Centered Design	301938.0000	9.01892e-241	Yes
Enneagram of Personality Traits	Erikson's Psychosocial Stages	323162.0000	4.96743e-249	Yes
Enneagram of Personality Traits	Flow Theory	306835.0000	9.97809e-243	Yes
Enneagram of Personality Traits	Freudian Psychoanalysis	0.0000	1.6501e-247	Yes
Enneagram of Personality Traits	Mental Models	325977.0000	4.06852e-250	Yes
Enneagram of Personality Traits	Myers-Briggs Type Indicator	323725.0000	3.01157e-249	Yes
Enneagram of Personality Traits	Social Cognitive Theory	308524.0000	2.22338e-243	Yes
Enneagram of Personality Traits	User Design Persona	0.0000	8.19353e-249	Yes
Enneagram of Personality Traits	User-Centered Design	311339.0000	1.82095e-244	Yes
Erikson's Psychosocial Stages	Flow Theory	312830.0000	4.0576e-245	Yes
Erikson's Psychosocial Stages	Freudian Psychoanalysis	0.0000	6.71082e-250	Yes
Erikson's Psychosocial Stages	Mental Models	332346.0000	1.65471e-252	Yes
Erikson's Psychosocial Stages	Myers-Briggs Type Indicator	330050.0000	1.22482e-251	Yes
Erikson's Psychosocial Stages	Social Cognitive Theory	314552.0000	9.04155e-246	Yes
Erikson's Psychosocial Stages	User Design Persona	0.0000	3.33231e-251	Yes
Erikson's Psychosocial Stages	User-Centered Design	317422.0000	7.40516e-247	Yes
Flow Theory	Freudian Psychoanalysis	0.0000	1.34794e-243	Yes
Flow Theory	Mental Models	315555.0000	3.32322e-246	Yes
Flow Theory	Myers-Briggs Type Indicator	0.0000	2.45996e-245	Yes
Flow Theory	Social Cognitive Theory	298660.0000	1.81649e-239	Yes
Flow Theory	User Design Persona	0.0000	6.69287e-245	Yes
Flow Theory	User-Centered Design	301385.0000	1.48765e-240	Yes
Freudian Psychoanalysis	Mental Models	328293.0000	5.49647e-251	Yes
Freudian Psychoanalysis	Myers-Briggs Type Indicator	326025.0000	4.06852e-250	Yes
Freudian Psychoanalysis	Social Cognitive Theory	310716.0000	3.00359e-244	Yes
Freudian Psychoanalysis	User Design Persona	0.0000	1.10691e-249	Yes
Freudian Psychoanalysis	User-Centered Design	313551.0000	2.45995e-245	Yes
Mental Models	Myers-Briggs Type Indicator	0.0000	1.0032e-252	Yes
Mental Models	Social Cognitive Theory	0.0000	7.40517e-247	Yes
Mental Models	User Design Persona	0.0000	2.72935e-252	Yes
Mental Models	User-Centered Design	320187.0000	6.06501e-248	Yes
Myers-Briggs Type Indicator	Social Cognitive Theory	315100.0000	5.48152e-246	Yes
Myers-Briggs Type Indicator	User Design Persona	0.0000	2.02027e-251	Yes
Myers-Briggs Type Indicator	User-Centered Design	317975.0000	4.48946e-247	Yes
Social Cognitive Theory	User Design Persona	0.0000	1.49137e-245	Yes
Social Cognitive Theory	User-Centered Design	303044.0000	3.31484e-241	Yes
User Design Persona	User-Centered Design	316869.0000	1.22145e-246	Yes

Table 12: Pairwise Weighted Mann–Whitney comparisons of description frameworks for Common Sense QA (Majority Vote).

Weighted Mann-Whitney for Final Decision Maker, Dataset: Common Sense QA				
Framework 1	Framework 2	MW-stat	p-value	Sig?
Big Five Personality Traits	Cognitive Behavioral Theory	318047.0000	4.48945e-247	Yes
Big Five Personality Traits	Cognitive Load Theory	0.0000	1.10396e-244	Yes
Big Five Personality Traits	Dual-Process Theory	304122.0000	1.21835e-241	Yes
Big Five Personality Traits	Enneagram of Personality Traits	0.0000	2.45995e-245	Yes
Big Five Personality Traits	Erikson's Psychosocial Stages	319718.0000	1.00039e-247	Yes
Big Five Personality Traits	Flow Theory	303565.0000	2.00063e-241	Yes
Big Five Personality Traits	Freudian Psychoanalysis	315819.0000	3.32321e-246	Yes
Big Five Personality Traits	Mental Models	322503.0000	8.19354e-249	Yes
Big Five Personality Traits	Myers-Briggs Type Indicator	320275.0000	6.065e-248	Yes
Big Five Personality Traits	Social Cognitive Theory	305236.0000	4.47797e-242	Yes
Big Five Personality Traits	User Design Persona	0.0000	1.65011e-247	Yes
Big Five Personality Traits	User-Centered Design	308021.0000	3.6674e-243	Yes
Cognitive Behavioral Theory	Cognitive Load Theory	0.0000	1.00039e-247	Yes
Cognitive Behavioral Theory	Dual-Process Theory	0.0000	1.10396e-244	Yes
Cognitive Behavioral Theory	Enneagram of Personality Traits	0.0000	2.22921e-248	Yes
Cognitive Behavioral Theory	Erikson's Psychosocial Stages	0.0000	9.06613e-251	Yes
Cognitive Behavioral Theory	Flow Theory	0.0000	1.82095e-244	Yes
Cognitive Behavioral Theory	Freudian Psychoanalysis	0.0000	3.01157e-249	Yes
Cognitive Behavioral Theory	Mental Models	330609.0000	7.42564e-252	Yes
Cognitive Behavioral Theory	Myers-Briggs Type Indicator	0.0000	5.49647e-251	Yes
Cognitive Behavioral Theory	Social Cognitive Theory	0.0000	4.0576e-245	Yes
Cognitive Behavioral Theory	User Design Persona	0.0000	1.49541e-250	Yes
Cognitive Behavioral Theory	User-Centered Design	0.0000	3.32322e-246	Yes
Cognitive Load Theory	Dual-Process Theory	305760.0000	2.71479e-242	Yes
Cognitive Load Theory	Enneagram of Personality Traits	0.0000	5.48151e-246	Yes
Cognitive Load Theory	Erikson's Psychosocial Stages	321440.0000	2.22921e-248	Yes
Cognitive Load Theory	Flow Theory	305200.0000	4.47797e-242	Yes
Cognitive Load Theory	Freudian Psychoanalysis	317520.0000	7.40516e-247	Yes
Cognitive Load Theory	Mental Models	324240.0000	1.8258e-249	Yes
Cognitive Load Theory	Myers-Briggs Type Indicator	322000.0000	1.35148e-248	Yes
Cognitive Load Theory	Social Cognitive Theory	306880.0000	9.97808e-243	Yes
Cognitive Load Theory	User Design Persona	0.0000	3.67698e-248	Yes
Cognitive Load Theory	User-Centered Design	309680.0000	8.17197e-244	Yes
Dual-Process Theory	Enneagram of Personality Traits	0.0000	6.04927e-243	Yes
Dual-Process Theory	Erikson's Psychosocial Stages	0.0000	2.45996e-245	Yes
Dual-Process Theory	Flow Theory	0.0000	4.9423e-239	Yes
Dual-Process Theory	Freudian Psychoanalysis	0.0000	8.17198e-244	Yes
Dual-Process Theory	Mental Models	316134.0000	2.01474e-246	Yes
Dual-Process Theory	Myers-Briggs Type Indicator	0.0000	1.49137e-245	Yes
Dual-Process Theory	Social Cognitive Theory	0.0000	1.10125e-239	Yes
Dual-Process Theory	User Design Persona	0.0000	4.0576e-245	Yes
Dual-Process Theory	User-Centered Design	0.0000	9.01892e-241	Yes
Enneagram of Personality Traits	Erikson's Psychosocial Stages	323162.0000	4.96743e-249	Yes
Enneagram of Personality Traits	Flow Theory	306835.0000	9.97808e-243	Yes
Enneagram of Personality Traits	Freudian Psychoanalysis	319221.0000	1.65011e-247	Yes
Enneagram of Personality Traits	Mental Models	325977.0000	4.06852e-250	Yes
Enneagram of Personality Traits	Myers-Briggs Type Indicator	323725.0000	3.01157e-249	Yes
Enneagram of Personality Traits	Social Cognitive Theory	308524.0000	2.22338e-243	Yes
Enneagram of Personality Traits	User Design Persona	0.0000	8.19353e-249	Yes
Enneagram of Personality Traits	User-Centered Design	311339.0000	1.82095e-244	Yes
Erikson's Psychosocial Stages	Flow Theory	312830.0000	4.0576e-245	Yes
Erikson's Psychosocial Stages	Freudian Psychoanalysis	325458.0000	6.71082e-250	Yes
Erikson's Psychosocial Stages	Mental Models	332346.0000	1.65471e-252	Yes
Erikson's Psychosocial Stages	Myers-Briggs Type Indicator	330050.0000	1.22482e-251	Yes
Erikson's Psychosocial Stages	Social Cognitive Theory	314552.0000	9.04155e-246	Yes
Erikson's Psychosocial Stages	User Design Persona	0.0000	3.32321e-251	Yes
Erikson's Psychosocial Stages	User-Centered Design	317422.0000	7.40516e-247	Yes
Flow Theory	Freudian Psychoanalysis	309015.0000	1.34794e-243	Yes
Flow Theory	Mental Models	315555.0000	3.32322e-246	Yes
Flow Theory	Myers-Briggs Type Indicator	313375.0000	2.45996e-245	Yes
Flow Theory	Social Cognitive Theory	298660.0000	1.81649e-239	Yes
Flow Theory	User Design Persona	0.0000	6.69287e-245	Yes
Flow Theory	User-Centered Design	301385.0000	1.48765e-240	Yes
Freudian Psychoanalysis	Mental Models	328293.0000	5.49647e-251	Yes
Freudian Psychoanalysis	Myers-Briggs Type Indicator	326025.0000	4.06852e-250	Yes
Freudian Psychoanalysis	Social Cognitive Theory	310716.0000	3.00359e-244	Yes
Freudian Psychoanalysis	User Design Persona	0.0000	1.10691e-249	Yes
Freudian Psychoanalysis	User-Centered Design	313551.0000	2.45995e-245	Yes
Mental Models	Myers-Briggs Type Indicator	0.0000	1.0032e-252	Yes
Mental Models	Social Cognitive Theory	0.0000	7.40517e-247	Yes
Mental Models	User Design Persona	0.0000	2.72935e-252	Yes
Mental Models	User-Centered Design	0.0000	6.06501e-248	Yes
Myers-Briggs Type Indicator	Social Cognitive Theory	315100.0000	5.48152e-246	Yes
Myers-Briggs Type Indicator	User Design Persona	0.0000	2.02027e-251	Yes
Myers-Briggs Type Indicator	User-Centered Design	317975.0000	4.48946e-247	Yes
Social Cognitive Theory	User Design Persona	0.0000	1.49137e-245	Yes
Social Cognitive Theory	User-Centered Design	303044.0000	3.31484e-241	Yes
User Design Persona	User-Centered Design	316869.0000	1.22145e-246	Yes

Table 13: Pairwise Weighted Mann-Whitney comparisons of description frameworks for Common Sense QA (Final Decision Maker).

Weighted Mann–Whitney for Final Decision Maker, Dataset: Last Letter Concat				
Framework 1	Framework 2	MW-stat	p-value	Sig?
Big Five Personality Traits	Cognitive Behavioral Theory	250000.0000	2.97418e-219	Yes
Big Five Personality Traits	Cognitive Load Theory	250000.0000	2.97418e-219	Yes
Big Five Personality Traits	Dual-Process Theory	0.0000	2.97418e-219	Yes
Big Five Personality Traits	Enneagram of Personality Traits	250000.0000	2.97418e-219	Yes
Big Five Personality Traits	Erikson's Psychosocial Stages	250000.0000	2.97418e-219	Yes
Big Five Personality Traits	Flow Theory	250000.0000	2.97418e-219	Yes
Big Five Personality Traits	Freudian Psychoanalysis	0.0000	2.97418e-219	Yes
Big Five Personality Traits	Mental Models	125000.0000	1	No
Big Five Personality Traits	Myers-Briggs Type Indicator	0.0000	2.97418e-219	Yes
Big Five Personality Traits	Social Cognitive Theory	0.0000	2.97418e-219	Yes
Big Five Personality Traits	User Design Persona	0.0000	2.97418e-219	Yes
Big Five Personality Traits	User-Centered Design	250000.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	Cognitive Load Theory	250000.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	Dual-Process Theory	0.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	Enneagram of Personality Traits	250000.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	Erikson's Psychosocial Stages	250000.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	Flow Theory	250000.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	Freudian Psychoanalysis	0.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	Mental Models	0.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	Myers-Briggs Type Indicator	0.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	Social Cognitive Theory	0.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	User Design Persona	0.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	User-Centered Design	250000.0000	2.97418e-219	Yes
Cognitive Load Theory	Dual-Process Theory	0.0000	2.97418e-219	Yes
Cognitive Load Theory	Enneagram of Personality Traits	0.0000	2.97418e-219	Yes
Cognitive Load Theory	Erikson's Psychosocial Stages	250000.0000	2.97418e-219	Yes
Cognitive Load Theory	Flow Theory	0.0000	2.97418e-219	Yes
Cognitive Load Theory	Freudian Psychoanalysis	0.0000	2.97418e-219	Yes
Cognitive Load Theory	Mental Models	0.0000	2.97418e-219	Yes
Cognitive Load Theory	Myers-Briggs Type Indicator	0.0000	2.97418e-219	Yes
Cognitive Load Theory	Social Cognitive Theory	0.0000	2.97418e-219	Yes
Cognitive Load Theory	User Design Persona	0.0000	2.97418e-219	Yes
Cognitive Load Theory	User-Centered Design	250000.0000	2.97418e-219	Yes
Dual-Process Theory	Enneagram of Personality Traits	250000.0000	2.97418e-219	Yes
Dual-Process Theory	Erikson's Psychosocial Stages	250000.0000	2.97418e-219	Yes
Dual-Process Theory	Flow Theory	250000.0000	2.97418e-219	Yes
Dual-Process Theory	Freudian Psychoanalysis	250000.0000	2.97418e-219	Yes
Dual-Process Theory	Mental Models	250000.0000	2.97418e-219	Yes
Dual-Process Theory	Myers-Briggs Type Indicator	250000.0000	2.97418e-219	Yes
Dual-Process Theory	Social Cognitive Theory	250000.0000	2.97418e-219	Yes
Dual-Process Theory	User Design Persona	250000.0000	2.97418e-219	Yes
Dual-Process Theory	User-Centered Design	250000.0000	2.97418e-219	Yes
Enneagram of Personality Traits	Erikson's Psychosocial Stages	250000.0000	2.97418e-219	Yes
Enneagram of Personality Traits	Flow Theory	250000.0000	2.97418e-219	Yes
Enneagram of Personality Traits	Freudian Psychoanalysis	0.0000	2.97418e-219	Yes
Enneagram of Personality Traits	Mental Models	0.0000	2.97418e-219	Yes
Enneagram of Personality Traits	Myers-Briggs Type Indicator	0.0000	2.97418e-219	Yes
Enneagram of Personality Traits	Social Cognitive Theory	0.0000	2.97418e-219	Yes
Enneagram of Personality Traits	User Design Persona	0.0000	2.97418e-219	Yes
Enneagram of Personality Traits	User-Centered Design	250000.0000	2.97418e-219	Yes
Erikson's Psychosocial Stages	Flow Theory	0.0000	2.97418e-219	Yes
Erikson's Psychosocial Stages	Freudian Psychoanalysis	0.0000	2.97418e-219	Yes
Erikson's Psychosocial Stages	Mental Models	0.0000	2.97418e-219	Yes
Erikson's Psychosocial Stages	Myers-Briggs Type Indicator	0.0000	2.97418e-219	Yes
Erikson's Psychosocial Stages	Social Cognitive Theory	0.0000	2.97418e-219	Yes
Erikson's Psychosocial Stages	User Design Persona	0.0000	2.97418e-219	Yes
Erikson's Psychosocial Stages	User-Centered Design	0.0000	2.97418e-219	Yes
Flow Theory	Freudian Psychoanalysis	0.0000	2.97418e-219	Yes
Flow Theory	Mental Models	0.0000	2.97418e-219	Yes
Flow Theory	Myers-Briggs Type Indicator	0.0000	2.97418e-219	Yes
Flow Theory	Social Cognitive Theory	0.0000	2.97418e-219	Yes
Flow Theory	User Design Persona	0.0000	2.97418e-219	Yes
Flow Theory	User-Centered Design	250000.0000	2.97418e-219	Yes
Freudian Psychoanalysis	Mental Models	250000.0000	2.97418e-219	Yes
Freudian Psychoanalysis	Myers-Briggs Type Indicator	250000.0000	2.97418e-219	Yes
Freudian Psychoanalysis	Social Cognitive Theory	250000.0000	2.97418e-219	Yes
Freudian Psychoanalysis	User Design Persona	250000.0000	2.97418e-219	Yes
Freudian Psychoanalysis	User-Centered Design	250000.0000	2.97418e-219	Yes
Mental Models	Myers-Briggs Type Indicator	0.0000	2.97418e-219	Yes
Mental Models	Social Cognitive Theory	0.0000	2.97418e-219	Yes
Mental Models	User Design Persona	0.0000	2.97418e-219	Yes
Mental Models	User-Centered Design	250000.0000	2.97418e-219	Yes
Myers-Briggs Type Indicator	Social Cognitive Theory	250000.0000	2.97418e-219	Yes
Myers-Briggs Type Indicator	User Design Persona	125000.0000	1	No
Myers-Briggs Type Indicator	User-Centered Design	250000.0000	2.97418e-219	Yes
Social Cognitive Theory	User Design Persona	0.0000	2.97418e-219	Yes
Social Cognitive Theory	User-Centered Design	250000.0000	2.97418e-219	Yes
User Design Persona	User-Centered Design	250000.0000	2.97418e-219	Yes

Table 14: Pairwise Weighted Mann–Whitney comparisons of description frameworks for Last Letter Concat (Final Decision Maker).

Weighted Mann–Whitney for Majority Vote, Dataset: Last Letters				
Framework 1	Framework 2	MW-stat	p-value	Sig?
Big Five Personality Traits	Cognitive Behavioral Theory	125000.0000	1	No
Big Five Personality Traits	Cognitive Load Theory	250000.0000	2.97418e-219	Yes
Big Five Personality Traits	Dual-Process Theory	0.0000	2.97418e-219	Yes
Big Five Personality Traits	Enneagram of Personality Traits	250000.0000	2.97418e-219	Yes
Big Five Personality Traits	Erikson's Psychosocial Stages	250000.0000	2.97418e-219	Yes
Big Five Personality Traits	Flow Theory	250000.0000	2.97418e-219	Yes
Big Five Personality Traits	Freudian Psychoanalysis	0.0000	2.97418e-219	Yes
Big Five Personality Traits	Mental Models	250000.0000	2.97418e-219	Yes
Big Five Personality Traits	Myers-Briggs Type Indicator	0.0000	2.97418e-219	Yes
Big Five Personality Traits	Social Cognitive Theory	250000.0000	2.97418e-219	Yes
Big Five Personality Traits	User Design Persona	0.0000	2.97418e-219	Yes
Big Five Personality Traits	User-Centered Design	250000.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	Cognitive Load Theory	250000.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	Dual-Process Theory	0.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	Enneagram of Personality Traits	250000.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	Erikson's Psychosocial Stages	250000.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	Flow Theory	250000.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	Freudian Psychoanalysis	0.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	Mental Models	250000.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	Myers-Briggs Type Indicator	0.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	Social Cognitive Theory	250000.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	User Design Persona	0.0000	2.97418e-219	Yes
Cognitive Behavioral Theory	User-Centered Design	250000.0000	2.97418e-219	Yes
Cognitive Load Theory	Dual-Process Theory	0.0000	2.97418e-219	Yes
Cognitive Load Theory	Enneagram of Personality Traits	0.0000	2.97418e-219	Yes
Cognitive Load Theory	Erikson's Psychosocial Stages	250000.0000	2.97418e-219	Yes
Cognitive Load Theory	Flow Theory	0.0000	2.97418e-219	Yes
Cognitive Load Theory	Freudian Psychoanalysis	0.0000	2.97418e-219	Yes
Cognitive Load Theory	Mental Models	0.0000	2.97418e-219	Yes
Cognitive Load Theory	Myers-Briggs Type Indicator	0.0000	2.97418e-219	Yes
Cognitive Load Theory	Social Cognitive Theory	250000.0000	2.97418e-219	Yes
Cognitive Load Theory	User Design Persona	0.0000	2.97418e-219	Yes
Cognitive Load Theory	User-Centered Design	250000.0000	2.97418e-219	Yes
Dual-Process Theory	Enneagram of Personality Traits	250000.0000	2.97418e-219	Yes
Dual-Process Theory	Erikson's Psychosocial Stages	250000.0000	2.97418e-219	Yes
Dual-Process Theory	Flow Theory	250000.0000	2.97418e-219	Yes
Dual-Process Theory	Freudian Psychoanalysis	250000.0000	2.97418e-219	Yes
Dual-Process Theory	Mental Models	250000.0000	2.97418e-219	Yes
Dual-Process Theory	Myers-Briggs Type Indicator	250000.0000	2.97418e-219	Yes
Dual-Process Theory	Social Cognitive Theory	250000.0000	2.97418e-219	Yes
Dual-Process Theory	User Design Persona	250000.0000	2.97418e-219	Yes
Dual-Process Theory	User-Centered Design	250000.0000	2.97418e-219	Yes
Enneagram of Personality Traits	Erikson's Psychosocial Stages	250000.0000	2.97418e-219	Yes
Enneagram of Personality Traits	Flow Theory	0.0000	2.97418e-219	Yes
Enneagram of Personality Traits	Freudian Psychoanalysis	0.0000	2.97418e-219	Yes
Enneagram of Personality Traits	Mental Models	0.0000	2.97418e-219	Yes
Enneagram of Personality Traits	Myers-Briggs Type Indicator	0.0000	2.97418e-219	Yes
Enneagram of Personality Traits	Social Cognitive Theory	250000.0000	2.97418e-219	Yes
Enneagram of Personality Traits	User Design Persona	0.0000	2.97418e-219	Yes
Enneagram of Personality Traits	User-Centered Design	250000.0000	2.97418e-219	Yes
Erikson's Psychosocial Stages	Flow Theory	0.0000	2.97418e-219	Yes
Erikson's Psychosocial Stages	Freudian Psychoanalysis	0.0000	2.97418e-219	Yes
Erikson's Psychosocial Stages	Mental Models	0.0000	2.97418e-219	Yes
Erikson's Psychosocial Stages	Myers-Briggs Type Indicator	0.0000	2.97418e-219	Yes
Erikson's Psychosocial Stages	Social Cognitive Theory	250000.0000	2.97418e-219	Yes
Erikson's Psychosocial Stages	User Design Persona	0.0000	2.97418e-219	Yes
Erikson's Psychosocial Stages	User-Centered Design	250000.0000	2.97418e-219	Yes
Flow Theory	Freudian Psychoanalysis	0.0000	2.97418e-219	Yes
Flow Theory	Mental Models	0.0000	2.97418e-219	Yes
Flow Theory	Myers-Briggs Type Indicator	0.0000	2.97418e-219	Yes
Flow Theory	Social Cognitive Theory	250000.0000	2.97418e-219	Yes
Flow Theory	User Design Persona	0.0000	2.97418e-219	Yes
Flow Theory	User-Centered Design	250000.0000	2.97418e-219	Yes
Freudian Psychoanalysis	Mental Models	250000.0000	2.97418e-219	Yes
Freudian Psychoanalysis	Myers-Briggs Type Indicator	125000.0000	1	No
Freudian Psychoanalysis	Social Cognitive Theory	250000.0000	2.97418e-219	Yes
Freudian Psychoanalysis	User Design Persona	0.0000	2.97418e-219	Yes
Freudian Psychoanalysis	User-Centered Design	250000.0000	2.97418e-219	Yes
Mental Models	Myers-Briggs Type Indicator	0.0000	2.97418e-219	Yes
Mental Models	Social Cognitive Theory	250000.0000	2.97418e-219	Yes
Mental Models	User Design Persona	0.0000	2.97418e-219	Yes
Mental Models	User-Centered Design	250000.0000	2.97418e-219	Yes
Myers-Briggs Type Indicator	Social Cognitive Theory	250000.0000	2.97418e-219	Yes
Myers-Briggs Type Indicator	User Design Persona	0.0000	2.97418e-219	Yes
Myers-Briggs Type Indicator	User-Centered Design	250000.0000	2.97418e-219	Yes
Social Cognitive Theory	User Design Persona	0.0000	2.97418e-219	Yes
Social Cognitive Theory	User-Centered Design	250000.0000	2.97418e-219	Yes
User Design Persona	User-Centered Design	250000.0000	2.97418e-219	Yes

Table 15: Pairwise Weighted Mann–Whitney comparisons of description frameworks for Last Letter Concat (Majority Vote).

Weighted Mann–Whitney for Final Decision Maker, Dataset: SocialIqa				
Framework 1	Framework 2	MW-stat	p-value	Sig?
Big Five Personality Traits	Cognitive Behavioral Theory	0.0000	0	Yes
Big Five Personality Traits	Cognitive Load Theory	2794800.0000	0	Yes
Big Five Personality Traits	Dual-Process Theory	2762500.0000	0	Yes
Big Five Personality Traits	Enneagram of Personality Traits	2839000.0000	0	Yes
Big Five Personality Traits	Erikson's Psychosocial Stages	2952900.0000	0	Yes
Big Five Personality Traits	Flow Theory	2811800.0000	0	Yes
Big Five Personality Traits	Freudian Psychoanalysis	2927400.0000	0	Yes
Big Five Personality Traits	Mental Models	2907000.0000	0	Yes
Big Five Personality Traits	Myers-Briggs Type Indicator	2881500.0000	0	Yes
Big Five Personality Traits	Social Cognitive Theory	2759100.0000	0	Yes
Big Five Personality Traits	User Design Persona	0.0000	0	Yes
Big Five Personality Traits	User-Centered Design	2832200.0000	0	Yes
Cognitive Behavioral Theory	Cognitive Load Theory	2706024.0000	0	Yes
Cognitive Behavioral Theory	Dual-Process Theory	2674750.0000	0	Yes
Cognitive Behavioral Theory	Enneagram of Personality Traits	2748820.0000	0	Yes
Cognitive Behavioral Theory	Erikson's Psychosocial Stages	2859102.0000	0	Yes
Cognitive Behavioral Theory	Flow Theory	2722484.0000	0	Yes
Cognitive Behavioral Theory	Freudian Psychoanalysis	2834412.0000	0	Yes
Cognitive Behavioral Theory	Mental Models	2814060.0000	0	Yes
Cognitive Behavioral Theory	Myers-Briggs Type Indicator	2789970.0000	0	Yes
Cognitive Behavioral Theory	Social Cognitive Theory	2671458.0000	0	Yes
Cognitive Behavioral Theory	User Design Persona	0.0000	0	Yes
Cognitive Behavioral Theory	User-Centered Design	2742236.0000	0	Yes
Cognitive Load Theory	Dual-Process Theory	2671500.0000	0	Yes
Cognitive Load Theory	Enneagram of Personality Traits	0.0000	0	Yes
Cognitive Load Theory	Erikson's Psychosocial Stages	2855628.0000	0	Yes
Cognitive Load Theory	Flow Theory	0.0000	0	Yes
Cognitive Load Theory	Freudian Psychoanalysis	0.0000	0	Yes
Cognitive Load Theory	Mental Models	2811240.0000	0	Yes
Cognitive Load Theory	Myers-Briggs Type Indicator	2786580.0000	0	Yes
Cognitive Load Theory	Social Cognitive Theory	0.0000	0	Yes
Cognitive Load Theory	User Design Persona	0.0000	0	Yes
Cognitive Load Theory	User-Centered Design	2738904.0000	0	Yes
Dual-Process Theory	Enneagram of Personality Traits	0.0000	0	Yes
Dual-Process Theory	Erikson's Psychosocial Stages	0.0000	0	Yes
Dual-Process Theory	Flow Theory	0.0000	0	Yes
Dual-Process Theory	Freudian Psychoanalysis	0.0000	0	Yes
Dual-Process Theory	Mental Models	0.0000	0	Yes
Dual-Process Theory	Myers-Briggs Type Indicator	2754375.0000	0	Yes
Dual-Process Theory	Social Cognitive Theory	0.0000	0	Yes
Dual-Process Theory	User Design Persona	0.0000	0	Yes
Dual-Process Theory	User-Centered Design	0.0000	0	Yes
Enneagram of Personality Traits	Erikson's Psychosocial Stages	2900790.0000	0	Yes
Enneagram of Personality Traits	Flow Theory	0.0000	0	Yes
Enneagram of Personality Traits	Freudian Psychoanalysis	0.0000	0	Yes
Enneagram of Personality Traits	Mental Models	2855700.0000	0	Yes
Enneagram of Personality Traits	Myers-Briggs Type Indicator	2830650.0000	0	Yes
Enneagram of Personality Traits	Social Cognitive Theory	0.0000	0	Yes
Enneagram of Personality Traits	User Design Persona	0.0000	0	Yes
Enneagram of Personality Traits	User-Centered Design	2782220.0000	0	Yes
Erikson's Psychosocial Stages	Flow Theory	0.0000	0	Yes
Erikson's Psychosocial Stages	Freudian Psychoanalysis	0.0000	0	Yes
Erikson's Psychosocial Stages	Mental Models	2970270.0000	0	Yes
Erikson's Psychosocial Stages	Myers-Briggs Type Indicator	2944215.0000	0	Yes
Erikson's Psychosocial Stages	Social Cognitive Theory	0.0000	0	Yes
Erikson's Psychosocial Stages	User Design Persona	0.0000	0	Yes
Erikson's Psychosocial Stages	User-Centered Design	2893842.0000	0	Yes
Flow Theory	Freudian Psychoanalysis	0.0000	0	Yes
Flow Theory	Mental Models	2828340.0000	0	Yes
Flow Theory	Myers-Briggs Type Indicator	2803530.0000	0	Yes
Flow Theory	Social Cognitive Theory	2684442.0000	0	Yes
Flow Theory	User Design Persona	0.0000	0	Yes
Flow Theory	User-Centered Design	2755564.0000	0	Yes
Freudian Psychoanalysis	Mental Models	2944620.0000	0	Yes
Freudian Psychoanalysis	Myers-Briggs Type Indicator	2918790.0000	0	Yes
Freudian Psychoanalysis	Social Cognitive Theory	2794806.0000	0	Yes
Freudian Psychoanalysis	User Design Persona	0.0000	0	Yes
Freudian Psychoanalysis	User-Centered Design	2868852.0000	0	Yes
Mental Models	Myers-Briggs Type Indicator	2898450.0000	0	Yes
Mental Models	Social Cognitive Theory	0.0000	0	Yes
Mental Models	User Design Persona	0.0000	0	Yes
Mental Models	User-Centered Design	2848860.0000	0	Yes
Myers-Briggs Type Indicator	Social Cognitive Theory	0.0000	0	Yes
Myers-Briggs Type Indicator	User Design Persona	0.0000	0	Yes
Myers-Briggs Type Indicator	User-Centered Design	0.0000	0	Yes
Social Cognitive Theory	User Design Persona	0.0000	0	Yes
Social Cognitive Theory	User-Centered Design	2703918.0000	0	Yes
User Design Persona	User-Centered Design	2857190.0000	0	Yes

Table 16: Pairwise Weighted Mann–Whitney comparisons of description frameworks for Social IQa (Final Decision Maker).

Weighted Mann–Whitney for Majority Vote, Dataset: SocialIQa				
Framework 1	Framework 2	MW-stat	p-value	Sig?
Big Five Personality Traits	Cognitive Behavioral Theory	0.0000	0	Yes
Big Five Personality Traits	Cognitive Load Theory	2794800.0000	0	Yes
Big Five Personality Traits	Dual-Process Theory	2762500.0000	0	Yes
Big Five Personality Traits	Enneagram of Personality Traits	2839000.0000	0	Yes
Big Five Personality Traits	Erikson's Psychosocial Stages	2952900.0000	0	Yes
Big Five Personality Traits	Flow Theory	2811800.0000	0	Yes
Big Five Personality Traits	Freudian Psychoanalysis	2927400.0000	0	Yes
Big Five Personality Traits	Mental Models	2907000.0000	0	Yes
Big Five Personality Traits	Myers-Briggs Type Indicator	2881500.0000	0	Yes
Big Five Personality Traits	Social Cognitive Theory	2759100.0000	0	Yes
Big Five Personality Traits	User Design Persona	0.0000	0	Yes
Big Five Personality Traits	User-Centered Design	2832200.0000	0	Yes
Cognitive Behavioral Theory	Cognitive Load Theory	2706024.0000	0	Yes
Cognitive Behavioral Theory	Dual-Process Theory	2674750.0000	0	Yes
Cognitive Behavioral Theory	Enneagram of Personality Traits	2748820.0000	0	Yes
Cognitive Behavioral Theory	Erikson's Psychosocial Stages	2859102.0000	0	Yes
Cognitive Behavioral Theory	Flow Theory	2722484.0000	0	Yes
Cognitive Behavioral Theory	Freudian Psychoanalysis	2834412.0000	0	Yes
Cognitive Behavioral Theory	Mental Models	2814060.0000	0	Yes
Cognitive Behavioral Theory	Myers-Briggs Type Indicator	2789970.0000	0	Yes
Cognitive Behavioral Theory	Social Cognitive Theory	2671458.0000	0	Yes
Cognitive Behavioral Theory	User Design Persona	0.0000	0	Yes
Cognitive Behavioral Theory	User-Centered Design	2742236.0000	0	Yes
Cognitive Load Theory	Dual-Process Theory	0.0000	0	Yes
Cognitive Load Theory	Enneagram of Personality Traits	0.0000	0	Yes
Cognitive Load Theory	Erikson's Psychosocial Stages	2855628.0000	0	Yes
Cognitive Load Theory	Flow Theory	0.0000	0	Yes
Cognitive Load Theory	Freudian Psychoanalysis	0.0000	0	Yes
Cognitive Load Theory	Mental Models	2811240.0000	0	Yes
Cognitive Load Theory	Myers-Briggs Type Indicator	2786580.0000	0	Yes
Cognitive Load Theory	Social Cognitive Theory	0.0000	0	Yes
Cognitive Load Theory	User Design Persona	0.0000	0	Yes
Cognitive Load Theory	User-Centered Design	0.0000	0	Yes
Dual-Process Theory	Enneagram of Personality Traits	0.0000	0	Yes
Dual-Process Theory	Erikson's Psychosocial Stages	2822625.0000	0	Yes
Dual-Process Theory	Flow Theory	1343875.0000	1	No
Dual-Process Theory	Freudian Psychoanalysis	2798250.0000	0	Yes
Dual-Process Theory	Mental Models	2778750.0000	0	Yes
Dual-Process Theory	Myers-Briggs Type Indicator	2754375.0000	0	Yes
Dual-Process Theory	Social Cognitive Theory	0.0000	0	Yes
Dual-Process Theory	User Design Persona	0.0000	0	Yes
Dual-Process Theory	User-Centered Design	2707250.0000	0	Yes
Enneagram of Personality Traits	Erikson's Psychosocial Stages	2900790.0000	0	Yes
Enneagram of Personality Traits	Flow Theory	2762180.0000	0	Yes
Enneagram of Personality Traits	Freudian Psychoanalysis	2875740.0000	0	Yes
Enneagram of Personality Traits	Mental Models	2855700.0000	0	Yes
Enneagram of Personality Traits	Myers-Briggs Type Indicator	2830650.0000	0	Yes
Enneagram of Personality Traits	Social Cognitive Theory	0.0000	0	Yes
Enneagram of Personality Traits	User Design Persona	0.0000	0	Yes
Enneagram of Personality Traits	User-Centered Design	2782220.0000	0	Yes
Erikson's Psychosocial Stages	Flow Theory	0.0000	0	Yes
Erikson's Psychosocial Stages	Freudian Psychoanalysis	0.0000	0	Yes
Erikson's Psychosocial Stages	Mental Models	2970270.0000	0	Yes
Erikson's Psychosocial Stages	Myers-Briggs Type Indicator	2944215.0000	0	Yes
Erikson's Psychosocial Stages	Social Cognitive Theory	0.0000	0	Yes
Erikson's Psychosocial Stages	User Design Persona	0.0000	0	Yes
Erikson's Psychosocial Stages	User-Centered Design	0.0000	0	Yes
Flow Theory	Freudian Psychoanalysis	2848188.0000	0	Yes
Flow Theory	Mental Models	2828340.0000	0	Yes
Flow Theory	Myers-Briggs Type Indicator	2803530.0000	0	Yes
Flow Theory	Social Cognitive Theory	0.0000	0	Yes
Flow Theory	User Design Persona	0.0000	0	Yes
Flow Theory	User-Centered Design	2755564.0000	0	Yes
Freudian Psychoanalysis	Mental Models	2944620.0000	0	Yes
Freudian Psychoanalysis	Myers-Briggs Type Indicator	2918790.0000	0	Yes
Freudian Psychoanalysis	Social Cognitive Theory	0.0000	0	Yes
Freudian Psychoanalysis	User Design Persona	0.0000	0	Yes
Freudian Psychoanalysis	User-Centered Design	0.0000	0	Yes
Mental Models	Myers-Briggs Type Indicator	0.0000	0	Yes
Mental Models	Social Cognitive Theory	0.0000	0	Yes
Mental Models	User Design Persona	0.0000	0	Yes
Mental Models	User-Centered Design	0.0000	0	Yes
Myers-Briggs Type Indicator	Social Cognitive Theory	0.0000	0	Yes
Myers-Briggs Type Indicator	User Design Persona	0.0000	0	Yes
Myers-Briggs Type Indicator	User-Centered Design	0.0000	0	Yes
Social Cognitive Theory	User Design Persona	0.0000	0	Yes
Social Cognitive Theory	User-Centered Design	2703918.0000	0	Yes
User Design Persona	User-Centered Design	2857190.0000	0	Yes

Table 17: Pairwise Weighted Mann–Whitney comparisons of description frameworks for Social IQa (Majority Vote).

Weighted Mann–Whitney for Final Decision Maker, Dataset: Social Support					
Framework 1	Framework 2	MW-stat	p-value	Sig?	
Big Five Personality Traits	Cognitive Behavioral Theory	0.0000	0	Yes	
Big Five Personality Traits	Cognitive Load Theory	402304.5000	1	No	
Big Five Personality Traits	Dual-Process Theory	0.0000	0	Yes	
Big Five Personality Traits	Enneagram of Personality Traits	0.0000	0	Yes	
Big Five Personality Traits	Erikson's Psychosocial Stages	0.0000	0	Yes	
Big Five Personality Traits	Flow Theory	0.0000	0	Yes	
Big Five Personality Traits	Freudian Psychoanalysis	0.0000	0	Yes	
Big Five Personality Traits	Mental Models	0.0000	0	Yes	
Big Five Personality Traits	Myers-Briggs Type Indicator	0.0000	0	Yes	
Big Five Personality Traits	Social Cognitive Theory	0.0000	0	Yes	
Big Five Personality Traits	User Design Persona	0.0000	0	Yes	
Big Five Personality Traits	User-Centered Design	0.0000	0	Yes	
Cognitive Behavioral Theory	Cognitive Load Theory	804609.0000	0	Yes	
Cognitive Behavioral Theory	Dual-Process Theory	804609.0000	0	Yes	
Cognitive Behavioral Theory	Enneagram of Personality Traits	804609.0000	0	Yes	
Cognitive Behavioral Theory	Erikson's Psychosocial Stages	804609.0000	0	Yes	
Cognitive Behavioral Theory	Flow Theory	804609.0000	0	Yes	
Cognitive Behavioral Theory	Freudian Psychoanalysis	804609.0000	0	Yes	
Cognitive Behavioral Theory	Mental Models	804609.0000	0	Yes	
Cognitive Behavioral Theory	Myers-Briggs Type Indicator	804609.0000	0	Yes	
Cognitive Behavioral Theory	Social Cognitive Theory	402304.5000	1	No	
Cognitive Behavioral Theory	User Design Persona	804609.0000	0	Yes	
Cognitive Behavioral Theory	User-Centered Design	804609.0000	0	Yes	
Cognitive Load Theory	Dual-Process Theory	0.0000	0	Yes	
Cognitive Load Theory	Enneagram of Personality Traits	0.0000	0	Yes	
Cognitive Load Theory	Erikson's Psychosocial Stages	0.0000	0	Yes	
Cognitive Load Theory	Flow Theory	0.0000	0	Yes	
Cognitive Load Theory	Freudian Psychoanalysis	0.0000	0	Yes	
Cognitive Load Theory	Mental Models	0.0000	0	Yes	
Cognitive Load Theory	Myers-Briggs Type Indicator	0.0000	0	Yes	
Cognitive Load Theory	Social Cognitive Theory	0.0000	0	Yes	
Cognitive Load Theory	User Design Persona	0.0000	0	Yes	
Cognitive Load Theory	User-Centered Design	0.0000	0	Yes	
Dual-Process Theory	Enneagram of Personality Traits	0.0000	0	Yes	
Dual-Process Theory	Erikson's Psychosocial Stages	804609.0000	0	Yes	
Dual-Process Theory	Flow Theory	0.0000	0	Yes	
Dual-Process Theory	Freudian Psychoanalysis	0.0000	0	Yes	
Dual-Process Theory	Mental Models	0.0000	0	Yes	
Dual-Process Theory	Myers-Briggs Type Indicator	0.0000	0	Yes	
Dual-Process Theory	Social Cognitive Theory	0.0000	0	Yes	
Dual-Process Theory	User Design Persona	804609.0000	0	Yes	
Dual-Process Theory	User-Centered Design	0.0000	0	Yes	
Enneagram of Personality Traits	Erikson's Psychosocial Stages	804609.0000	0	Yes	
Enneagram of Personality Traits	Flow Theory	804609.0000	0	Yes	
Enneagram of Personality Traits	Freudian Psychoanalysis	804609.0000	0	Yes	
Enneagram of Personality Traits	Mental Models	804609.0000	0	Yes	
Enneagram of Personality Traits	Myers-Briggs Type Indicator	0.0000	0	Yes	
Enneagram of Personality Traits	Social Cognitive Theory	0.0000	0	Yes	
Enneagram of Personality Traits	User Design Persona	804609.0000	0	Yes	
Enneagram of Personality Traits	User-Centered Design	0.0000	0	Yes	
Erikson's Psychosocial Stages	Flow Theory	0.0000	0	Yes	
Erikson's Psychosocial Stages	Freudian Psychoanalysis	0.0000	0	Yes	
Erikson's Psychosocial Stages	Mental Models	0.0000	0	Yes	
Erikson's Psychosocial Stages	Myers-Briggs Type Indicator	0.0000	0	Yes	
Erikson's Psychosocial Stages	Social Cognitive Theory	0.0000	0	Yes	
Erikson's Psychosocial Stages	User Design Persona	0.0000	0	Yes	
Erikson's Psychosocial Stages	User-Centered Design	0.0000	0	Yes	
Flow Theory	Freudian Psychoanalysis	804609.0000	0	Yes	
Flow Theory	Mental Models	804609.0000	0	Yes	
Flow Theory	Myers-Briggs Type Indicator	0.0000	0	Yes	
Flow Theory	Social Cognitive Theory	0.0000	0	Yes	
Flow Theory	User Design Persona	804609.0000	0	Yes	
Flow Theory	User-Centered Design	0.0000	0	Yes	
Freudian Psychoanalysis	Mental Models	0.0000	0	Yes	
Freudian Psychoanalysis	Myers-Briggs Type Indicator	0.0000	0	Yes	
Freudian Psychoanalysis	Social Cognitive Theory	0.0000	0	Yes	
Freudian Psychoanalysis	User Design Persona	804609.0000	0	Yes	
Freudian Psychoanalysis	User-Centered Design	0.0000	0	Yes	
Mental Models	Myers-Briggs Type Indicator	0.0000	0	Yes	
Mental Models	Social Cognitive Theory	0.0000	0	Yes	
Mental Models	User Design Persona	804609.0000	0	Yes	
Mental Models	User-Centered Design	0.0000	0	Yes	
Myers-Briggs Type Indicator	Social Cognitive Theory	0.0000	0	Yes	
Myers-Briggs Type Indicator	User Design Persona	804609.0000	0	Yes	
Myers-Briggs Type Indicator	User-Centered Design	0.0000	0	Yes	
Social Cognitive Theory	User Design Persona	804609.0000	0	Yes	
Social Cognitive Theory	User-Centered Design	804609.0000	0	Yes	
User Design Persona	User-Centered Design	0.0000	0	Yes	

Table 18: Pairwise Weighted Mann–Whitney comparisons of description frameworks for Social Support (Final Decision Maker).

Weighted Mann–Whitney for Majority Vote, Dataset: Social Support				
Framework 1	Framework 2	MW-stat	p-value	Sig?
Big Five Personality Traits	Cognitive Behavioral Theory	0.0000	0	Yes
Big Five Personality Traits	Cognitive Load Theory	0.0000	0	Yes
Big Five Personality Traits	Dual-Process Theory	0.0000	0	Yes
Big Five Personality Traits	Enneagram of Personality Traits	0.0000	0	Yes
Big Five Personality Traits	Erikson's Psychosocial Stages	0.0000	0	Yes
Big Five Personality Traits	Flow Theory	0.0000	0	Yes
Big Five Personality Traits	Freudian Psychoanalysis	0.0000	0	Yes
Big Five Personality Traits	Mental Models	0.0000	0	Yes
Big Five Personality Traits	Myers-Briggs Type Indicator	0.0000	0	Yes
Big Five Personality Traits	Social Cognitive Theory	0.0000	0	Yes
Big Five Personality Traits	User Design Persona	0.0000	0	Yes
Big Five Personality Traits	User-Centered Design	0.0000	0	Yes
Cognitive Behavioral Theory	Cognitive Load Theory	804609.0000	0	Yes
Cognitive Behavioral Theory	Dual-Process Theory	804609.0000	0	Yes
Cognitive Behavioral Theory	Enneagram of Personality Traits	402304.5000	1	No
Cognitive Behavioral Theory	Erikson's Psychosocial Stages	804609.0000	0	Yes
Cognitive Behavioral Theory	Flow Theory	402304.5000	1	No
Cognitive Behavioral Theory	Freudian Psychoanalysis	804609.0000	0	Yes
Cognitive Behavioral Theory	Mental Models	804609.0000	0	Yes
Cognitive Behavioral Theory	Myers-Briggs Type Indicator	804609.0000	0	Yes
Cognitive Behavioral Theory	Social Cognitive Theory	0.0000	0	Yes
Cognitive Behavioral Theory	User Design Persona	804609.0000	0	Yes
Cognitive Behavioral Theory	User-Centered Design	804609.0000	0	Yes
Cognitive Load Theory	Dual-Process Theory	804609.0000	0	Yes
Cognitive Load Theory	Enneagram of Personality Traits	0.0000	0	Yes
Cognitive Load Theory	Erikson's Psychosocial Stages	0.0000	0	Yes
Cognitive Load Theory	Flow Theory	0.0000	0	Yes
Cognitive Load Theory	Freudian Psychoanalysis	0.0000	0	Yes
Cognitive Load Theory	Mental Models	0.0000	0	Yes
Cognitive Load Theory	Myers-Briggs Type Indicator	0.0000	0	Yes
Cognitive Load Theory	Social Cognitive Theory	0.0000	0	Yes
Cognitive Load Theory	User Design Persona	0.0000	0	Yes
Cognitive Load Theory	User-Centered Design	0.0000	0	Yes
Dual-Process Theory	Enneagram of Personality Traits	0.0000	0	Yes
Dual-Process Theory	Erikson's Psychosocial Stages	0.0000	0	Yes
Dual-Process Theory	Flow Theory	0.0000	0	Yes
Dual-Process Theory	Freudian Psychoanalysis	0.0000	0	Yes
Dual-Process Theory	Mental Models	0.0000	0	Yes
Dual-Process Theory	Myers-Briggs Type Indicator	0.0000	0	Yes
Dual-Process Theory	Social Cognitive Theory	0.0000	0	Yes
Dual-Process Theory	User Design Persona	0.0000	0	Yes
Dual-Process Theory	User-Centered Design	0.0000	0	Yes
Enneagram of Personality Traits	Erikson's Psychosocial Stages	804609.0000	0	Yes
Enneagram of Personality Traits	Flow Theory	402304.5000	1	No
Enneagram of Personality Traits	Freudian Psychoanalysis	804609.0000	0	Yes
Enneagram of Personality Traits	Mental Models	804609.0000	0	Yes
Enneagram of Personality Traits	Myers-Briggs Type Indicator	804609.0000	0	Yes
Enneagram of Personality Traits	Social Cognitive Theory	0.0000	0	Yes
Enneagram of Personality Traits	User Design Persona	804609.0000	0	Yes
Enneagram of Personality Traits	User-Centered Design	804609.0000	0	Yes
Erikson's Psychosocial Stages	Flow Theory	0.0000	0	Yes
Erikson's Psychosocial Stages	Freudian Psychoanalysis	804609.0000	0	Yes
Erikson's Psychosocial Stages	Mental Models	0.0000	0	Yes
Erikson's Psychosocial Stages	Myers-Briggs Type Indicator	0.0000	0	Yes
Erikson's Psychosocial Stages	Social Cognitive Theory	0.0000	0	Yes
Erikson's Psychosocial Stages	User Design Persona	0.0000	0	Yes
Erikson's Psychosocial Stages	User-Centered Design	0.0000	0	Yes
Flow Theory	Freudian Psychoanalysis	804609.0000	0	Yes
Flow Theory	Mental Models	804609.0000	0	Yes
Flow Theory	Myers-Briggs Type Indicator	804609.0000	0	Yes
Flow Theory	Social Cognitive Theory	0.0000	0	Yes
Flow Theory	User Design Persona	804609.0000	0	Yes
Flow Theory	User-Centered Design	804609.0000	0	Yes
Freudian Psychoanalysis	Mental Models	0.0000	0	Yes
Freudian Psychoanalysis	Myers-Briggs Type Indicator	0.0000	0	Yes
Freudian Psychoanalysis	Social Cognitive Theory	0.0000	0	Yes
Freudian Psychoanalysis	User Design Persona	0.0000	0	Yes
Freudian Psychoanalysis	User-Centered Design	0.0000	0	Yes
Mental Models	Myers-Briggs Type Indicator	0.0000	0	Yes
Mental Models	Social Cognitive Theory	0.0000	0	Yes
Mental Models	User Design Persona	0.0000	0	Yes
Mental Models	User-Centered Design	0.0000	0	Yes
Myers-Briggs Type Indicator	Social Cognitive Theory	0.0000	0	Yes
Myers-Briggs Type Indicator	User Design Persona	0.0000	0	Yes
Myers-Briggs Type Indicator	User-Centered Design	804609.0000	0	Yes
Social Cognitive Theory	User Design Persona	804609.0000	0	Yes
Social Cognitive Theory	User-Centered Design	804609.0000	0	Yes
User Design Persona	User-Centered Design	804609.0000	0	Yes

Table 19: Pairwise Weighted Mann–Whitney comparisons of description frameworks for Social Support (Majority Vote).

Permutation test for Final Decision Maker, Dataset: Social Support				
Framework 1	Framework 2	Obs. diff.	p-value	Sig?
Big Five Personality Traits	Cognitive Behavioral Theory	0.07478720098940464	0.0	Yes
Big Five Personality Traits	Cognitive Load Theory	0.00018706392265849248	0.978	No
Big Five Personality Traits	Dual-Process Theory	0.004288782590265855	0.436	No
Big Five Personality Traits	Enneagram of Personality Traits	0.06990413270040416	0.0	Yes
Big Five Personality Traits	Erikson's Psychosocial Stages	0.001208191025936306	0.827	No
Big Five Personality Traits	Flow Theory	0.06977930844300786	0.0	Yes
Big Five Personality Traits	Freudian Psychoanalysis	0.06478344650610132	0.0	Yes
Big Five Personality Traits	Mental Models	0.06484863387346404	0.0	Yes
Big Five Personality Traits	Myers-Briggs Type Indicator	0.1738772620351862	0.0	Yes
Big Five Personality Traits	Social Cognitive Theory	0.014690781269527331	0.025	Yes
Big Five Personality Traits	User Design Persona	0.0028452629964489107	0.631	No
Big Five Personality Traits	User-Centered Design	0.07536115789968492	0.0	Yes
Cognitive Behavioral Theory	Cognitive Load Theory	0.07497426491206313	0.0	Yes
Cognitive Behavioral Theory	Dual-Process Theory	0.07049841839913878	0.0	Yes
Cognitive Behavioral Theory	Enneagram of Personality Traits	0.00488306828900481	0.473	No
Cognitive Behavioral Theory	Erikson's Psychosocial Stages	0.07357900996346833	0.0	Yes
Cognitive Behavioral Theory	Flow Theory	0.005007892546396775	0.561	No
Cognitive Behavioral Theory	Freudian Psychoanalysis	0.010003754483303318	0.311	No
Cognitive Behavioral Theory	Mental Models	0.0099385671159406	0.347	No
Cognitive Behavioral Theory	Myers-Briggs Type Indicator	0.09090006104578155	0.0	Yes
Cognitive Behavioral Theory	Social Cognitive Theory	0.06009641971987731	0.0	Yes
Cognitive Behavioral Theory	User Design Persona	0.07194193799295573	0.0	Yes
Cognitive Behavioral Theory	User-Centered Design	0.0005739569102802822	0.941	No
Cognitive Load Theory	Dual-Process Theory	0.004475846512924347	0.425	No
Cognitive Load Theory	Enneagram of Personality Traits	0.07009119662306265	0.001	Yes
Cognitive Load Theory	Erikson's Psychosocial Stages	0.0013952549485947985	0.822	No
Cognitive Load Theory	Flow Theory	0.06996637236506636	0.0	Yes
Cognitive Load Theory	Freudian Psychoanalysis	0.06497051042875981	0.0	Yes
Cognitive Load Theory	Mental Models	0.06503569779612253	0.0	Yes
Cognitive Load Theory	Myers-Briggs Type Indicator	0.17406432595784468	0.0	Yes
Cognitive Load Theory	Social Cognitive Theory	0.014877845192185823	0.059	No
Cognitive Load Theory	User Design Persona	0.003032326919107403	0.561	No
Cognitive Load Theory	User-Centered Design	0.07554822182234341	0.0	Yes
Dual-Process Theory	Enneagram of Personality Traits	0.0656153501101383	0.0	Yes
Dual-Process Theory	Erikson's Psychosocial Stages	0.003080591564329549	0.566	No
Dual-Process Theory	Flow Theory	0.065490522585274201	0.0	Yes
Dual-Process Theory	Freudian Psychoanalysis	0.06049466391583547	0.001	Yes
Dual-Process Theory	Mental Models	0.060559851283198185	0.001	Yes
Dual-Process Theory	Myers-Briggs Type Indicator	0.10958847944492034	0.0	Yes
Dual-Process Theory	Social Cognitive Theory	0.010401998679261476	0.191	No
Dual-Process Theory	User Design Persona	0.0014455195938169443	0.797	No
Dual-Process Theory	User-Centered Design	0.07107237530941907	0.0	Yes
Enneagram of Personality Traits	Erikson's Psychosocial Stages	0.06869594167446785	0.001	Yes
Enneagram of Personality Traits	Flow Theory	0.00012482425739629432	0.987	No
Enneagram of Personality Traits	Freudian Psychoanalysis	0.005129686194302837	0.607	No
Enneagram of Personality Traits	Mental Models	0.005055498826940119	0.587	No
Enneagram of Personality Traits	Myers-Briggs Type Indicator	0.10397312933478203	0.0	Yes
Enneagram of Personality Traits	Social Cognitive Theory	0.05521335143087683	0.009	Yes
Enneagram of Personality Traits	User Design Persona	0.06705886970395525	0.0	Yes
Enneagram of Personality Traits	User-Centered Design	0.005457025199280763	0.5	No
Erikson's Psychosocial Stages	Flow Theory	0.06857111741707156	0.0	Yes
Erikson's Psychosocial Stages	Freudian Psychoanalysis	0.06357525548016502	0.0	Yes
Erikson's Psychosocial Stages	Mental Models	0.06364044284752773	0.0	Yes
Erikson's Psychosocial Stages	Myers-Briggs Type Indicator	0.17266907100924989	0.0	Yes
Erikson's Psychosocial Stages	Social Cognitive Theory	0.013482590243591025	0.089	No
Erikson's Psychosocial Stages	User Design Persona	0.0016370719705126047	0.73	No
Erikson's Psychosocial Stages	User-Centered Design	0.07415296687374862	0.0	Yes
Flow Theory	Freudian Psychoanalysis	0.004995861936906543	0.489	No
Flow Theory	Mental Models	0.004930674569543825	0.461	No
Flow Theory	Myers-Briggs Type Indicator	0.10409795359217833	0.0	Yes
Flow Theory	Social Cognitive Theory	0.055088527173480534	0.001	Yes
Flow Theory	User Design Persona	0.06693404544655895	0.0	Yes
Flow Theory	User-Centered Design	0.005581849456677057	0.394	No
Freudian Psychoanalysis	Mental Models	6.518736736271791e-05	0.984	No
Freudian Psychoanalysis	Myers-Briggs Type Indicator	0.10909381552908487	0.0	Yes
Freudian Psychoanalysis	Social Cognitive Theory	0.05009296523657399	0.009	Yes
Freudian Psychoanalysis	User Design Persona	0.06193818350965241	0.0	Yes
Freudian Psychoanalysis	User-Centered Design	0.0105777113935836	0.165	No
Mental Models	Myers-Briggs Type Indicator	0.10902862816172215	0.0	Yes
Mental Models	Social Cognitive Theory	0.05015785260393671	0.002	Yes
Mental Models	User Design Persona	0.0620037087701513	0.0	Yes
Mental Models	User-Centered Design	0.010512524026220882	0.12	No
Myers-Briggs Type Indicator	Social Cognitive Theory	0.15918648076565886	0.0	Yes
Myers-Briggs Type Indicator	User Design Persona	0.17103199903873728	0.0	Yes
Myers-Briggs Type Indicator	User-Centered Design	0.09851610413550127	0.0	Yes
Social Cognitive Theory	User Design Persona	0.01184551827307842	0.167	No
Social Cognitive Theory	User-Centered Design	0.06067037663015759	0.0	Yes
User Design Persona	User-Centered Design	0.07251589490323601	0.0	Yes

Table 20: Pairwise Permutational Test comparisons of description frameworks for Social Support (Final Decision Maker).

<i>Dataset</i>	<i>F-statistic</i>	<i>p-value</i>
Strategy QA	4.233278	0.002217
Common Sense QA	0.653576	0.660868
Last Letter Concat	0.504182	0.872550
Social IQa	8.901237	0.000003
Social Support	2.380994	0.045282

Table 21: Weighted ANOVA (Welch’s ANOVA) Test Results Within Each Dataset.

<i>Dataset</i>	<i>H-statistic</i>	<i>p-value</i>
Strategy QA	6.695412	0.461267
Common Sense QA	5.464388	0.361867
Last Letter Concat	5.247787	0.874023
Social IQa	14.857131	0.094936
Social Support	14.967613	0.036417

Table 22: Weighted Kruskal-Wallis Test Results Within Each Dataset.

<i>Dataset</i>	<i>Expert Role</i>	<i>Count</i>
Social IQa	Behavioral Analysis	1
Social IQa	Behavioral Science	1
Social IQa	Cognitive Behavioral Analysis	1
Social IQa	Communication Studies	1
Social IQa	Emotional Intelligence	10
Social IQa	Interpersonal Communication	6
Social IQa	Interpersonal Development	1
Social IQa	Psychology	5
Social IQa	Social Psychology	9
Social IQa	Sociology	4
Common Sense QA	Artificial Intelligence	13
Common Sense QA	Cognitive Psychology	4
Common Sense QA	Cognitive Science	8
Common Sense QA	Knowledge Representation and Reasoning	1
Common Sense QA	Linguistics	1
Common Sense QA	Natural Language Processing	12
Last Letter Concat	Algorithmic Problem Solving	1
Last Letter Concat	Coding Theory	1
Last Letter Concat	Computational Linguistics	8
Last Letter Concat	Cryptography	4
Last Letter Concat	Linguistics	1
Last Letter Concat	Morphology	7
Last Letter Concat	Natural Language Processing	4
Last Letter Concat	Orthography	2
Last Letter Concat	Phonetics	7
Last Letter Concat	Phonology	2
Last Letter Concat	String Manipulation	2
Social Support	Artificial Intelligence and Machine Learning	1
Social Support	Computational Linguistics	4
Social Support	Linguistics	3
Social Support	Machine Learning	5
Social Support	Natural Language Processing	13
Social Support	Psycholinguistics	1
Social Support	Psychology	1
Social Support	Social Psychology	11
Strategy QA	Artificial Intelligence	12
Strategy QA	Cognitive Psychology	6
Strategy QA	Cognitive Science	6
Strategy QA	Formal Logic	5
Strategy QA	Logic and Formal Reasoning	3
Strategy QA	Machine Learning	1
Strategy QA	Mathematical Logic	2
Strategy QA	Natural Language Processing	4

Table 23: Expert Role Distribution Within Each Dataset by summing the number of times a role is used among the description frameworks.

	Group 1	Group 2	Mean Diff.	p-value	Lower	Upper	Significant
1	Artificial Intelligence	Cognitive Psychology	1408.1692	0.0052	260.0531	2556.2852	True
2	Artificial Intelligence	Cognitive Science	84.5792	1.0000	-1063.5369	1232.6952	False
3	Artificial Intelligence	Formal Logic	345.3855	0.9893	-876.8774	1567.6484	False
4	Artificial Intelligence	Logic and Formal Reasoning	2407.1308	0.0000	924.9194	3889.3423	True
5	Artificial Intelligence	Machine Learning	3558.8175	0.0002	1168.8234	5948.8116	True
6	Artificial Intelligence	Mathematical Logic	-725.5825	0.9132	-2479.3587	1028.1937	False
7	Artificial Intelligence	Natural Language Processing	1418.9850	0.0262	93.2548	2744.7152	True
8	Cognitive Psychology	Cognitive Science	-1323.5900	0.0507	-2649.3202	2.1402	False
9	Cognitive Psychology	Formal Logic	-1062.7837	0.2812	-2453.2212	327.6539	False
10	Cognitive Psychology	Logic and Formal Reasoning	998.9617	0.5704	-624.7196	2622.6429	False
11	Cognitive Psychology	Machine Learning	2150.6483	0.1444	-329.5658	4630.8624	False
12	Cognitive Psychology	Mathematical Logic	-2133.7517	0.0134	-4008.6173	-258.8860	True
13	Cognitive Psychology	Natural Language Processing	10.8158	1.0000	-1471.3956	1493.0273	False
14	Cognitive Science	Formal Logic	260.8063	0.9992	-1129.6312	1651.2439	False
15	Cognitive Science	Logic and Formal Reasoning	2322.5517	0.0004	698.8704	3946.2329	True
16	Cognitive Science	Machine Learning	3474.2383	0.0006	994.0242	5954.4524	True
17	Cognitive Science	Mathematical Logic	-810.1617	0.8929	-2685.0273	1064.7040	False
18	Cognitive Science	Natural Language Processing	1334.4058	0.1131	-147.8056	2816.6173	False
19	Formal Logic	Logic and Formal Reasoning	2061.7453	0.0050	384.8145	3738.6761	True
20	Formal Logic	Machine Learning	3213.4320	0.0029	698.0358	5728.8282	True
21	Formal Logic	Mathematical Logic	-1070.9680	0.6894	-2992.1336	850.1976	False
22	Formal Logic	Natural Language Processing	1073.5995	0.4024	-466.7598	2613.9588	False
23	Logic and Formal Reasoning	Machine Learning	1151.6867	0.8903	-1499.7737	3803.1471	False
24	Logic and Formal Reasoning	Mathematical Logic	-3132.7133	0.0002	-5228.8768	-1036.5498	True
25	Logic and Formal Reasoning	Natural Language Processing	-988.1458	0.6775	-2741.9220	765.6304	False
26	Machine Learning	Mathematical Logic	-4284.4000	0.0001	-7096.6985	-1472.1015	True
27	Machine Learning	Natural Language Processing	-2139.8325	0.1824	-4707.0980	427.4330	False
28	Mathematical Logic	Natural Language Processing	2144.5675	0.0242	155.9722	4133.1628	True

Table 24: Tukey's HSD Pairwise Comparison Results for Strategy QA. The columns "Upper" and "Lower" refer to the confidence interval bounds for the difference between the means of the two groups.

	Group 1	Group 2	Mean Diff.	p-value	Lower	Upper	Significant
1	Behavioral Analysis	Behavioral Science	8151.6300	0.0000	4760.6515	11542.6085	True
2	Behavioral Analysis	Cognitive Behavioral Analysis	8241.9300	0.0000	4850.9515	11632.9085	True
3	Behavioral Analysis	Communication Studies	9021.0300	0.0000	5630.0515	12412.0085	True
4	Behavioral Analysis	Emotional Intelligence	4399.5480	0.0000	1884.7310	6914.3650	True
5	Behavioral Analysis	Interpersonal Communication	4109.0417	0.0000	1519.1391	6698.9443	True
6	Behavioral Analysis	Interpersonal Development	11538.6900	0.0000	8147.7115	14929.6685	True
7	Behavioral Analysis	Psychology	4738.2040	0.0000	2111.5633	7364.8447	True
8	Behavioral Analysis	Social Psychology	5968.3100	0.0000	3440.8239	8495.7961	True
9	Behavioral Analysis	Sociology	5467.3250	0.0000	2786.5211	8148.1289	True

Table 25: Tukey's HSD Pairwise Comparison Results for Social IQa. The columns "Upper" and "Lower" refer to the confidence interval bounds for the difference between the means of the two groups.

	Group 1	Group 2	Mean Diff.	p-value	Lower	Upper	Significant
1	Artificial Intelligence and Machine Learning	Computational Linguistics	-1141.4325	0.9839	-5261.5507	2978.6857	False
2	Artificial Intelligence and Machine Learning	Linguistics	-2146.8200	0.7247	-6402.0597	2108.4197	False
3	Artificial Intelligence and Machine Learning	Machine Learning	-1365.2340	0.9525	-5402.1089	2671.6409	False
4	Artificial Intelligence and Machine Learning	Natural Language Processing	-2370.8400	0.4902	-6195.0964	1453.4164	False
5	Artificial Intelligence and Machine Learning	Psycholinguistics	-1139.1900	0.9961	-6350.7730	4072.3930	False
6	Artificial Intelligence and Machine Learning	Psychology	-1542.8400	0.9767	-6754.4230	3668.7430	False
7	Artificial Intelligence and Machine Learning	Social Psychology	-680.9045	0.9990	-4529.9137	3168.1046	False
8	Computational Linguistics	Linguistics	-1005.3875	0.9373	-3819.9640	1809.1890	False
9	Computational Linguistics	Machine Learning	-223.8015	1.0000	-2695.8724	2248.2694	False
10	Computational Linguistics	Natural Language Processing	-1229.4075	0.5648	-3336.4702	877.6552	False
11	Computational Linguistics	Psycholinguistics	2.2425	1.0000	-4117.8757	4122.3607	False
12	Computational Linguistics	Psychology	-401.4075	1.0000	-4521.5257	3718.7107	False
13	Computational Linguistics	Social Psychology	460.5280	0.9965	-1691.1336	2612.1895	False
14	Linguistics	Machine Learning	781.5860	0.9790	-1909.6639	3472.8359	False
15	Linguistics	Natural Language Processing	-224.0200	1.0000	-2584.4023	2136.3623	False
16	Linguistics	Psycholinguistics	1007.6300	0.9936	-3247.6097	5262.8697	False
17	Linguistics	Psychology	603.9800	0.9998	-3651.2597	4859.2197	False
18	Linguistics	Social Psychology	1465.9155	0.5088	-934.3635	3866.1944	False
19	Machine Learning	Natural Language Processing	-1005.6060	0.6973	-2944.8580	933.6460	False
20	Machine Learning	Psycholinguistics	226.0440	1.0000	-3810.8309	4262.9189	False
21	Machine Learning	Psychology	-177.6060	1.0000	-4214.4809	3859.2689	False
22	Machine Learning	Social Psychology	684.3295	0.9479	-1303.2903	2671.9492	False
23	Natural Language Processing	Psycholinguistics	1231.6500	0.9632	-2592.6064	5055.9064	False
24	Natural Language Processing	Psychology	828.0000	0.9963	-2996.2564	4652.2564	False
25	Natural Language Processing	Social Psychology	1689.9355	0.0197	180.2299	3199.6411	True
26	Psycholinguistics	Psychology	-403.6500	1.0000	-5615.2330	4807.9330	False
27	Psycholinguistics	Social Psychology	458.2855	0.9999	-3390.7237	4307.2946	False
28	Psychology	Social Psychology	861.9355	0.9954	-2987.0737	4710.9446	False

Table 26: Tukey’s HSD Pairwise Comparison Results for Social Support. The columns "Upper" and "Lower" refer to the confidence interval bounds for the difference between the means of the two groups.

Expert	Successes	Failures
Expert 0	49 607.86	18 485.14
Expert 1	49 797.19	18 295.81
Expert 2	49 820.12	18 272.88

Table 27: Global Contingency Table of Successes and Failures by Expert Position (from 0 to n).